Metrics for Scientific Discovery: Auditing Embedding Spaces for Novelty, Calibration, and Interpretability

P. C. Harris
Laboratory for Nuclear Science
Massachusetts Institute of Technology
Cambridge, MA 02139, USA
pcharris@mit.edu

Abstract

Self-supervised objectives with explicit distributional priors — LeJEPA and its sketched-isotropic-Gaussian regularizer (SIGReg) being the canonical example — promise embedding spaces whose *geometry*, not merely whose linear separability, carries information. This promise matters most in scientific settings, where the object of interest is by construction *unlabeled*: a new particle, a new galaxy morphology, a new material phase. Yet the field evaluates such spaces almost exclusively with linear probes, which measure exactly the one property a discovery pipeline cannot rely on. We argue that a space intended for discovery must be audited in four distinct currencies — *detection*, *calibration*, *discoverability*, and *interpretability* — and we supply a concrete, reproducible battery for each. Our central result concerns the discovery of genuinely new classes: under a specified Gaussian marginal, such a pipeline is not three independent design choices but **one likelihood model at three scales**. The Neyman–Pearson density-ratio functional that shapes the embedding reappears verbatim as the SparKer detection statistic, and the enforced marginal is precisely what makes Euclidean k-means and a unit-variance BIC correctly specified for proposing the new classes — so representation, clustering, and hypothesis test read one quadratic surface. We characterize when this pipeline finds real classes (a *purity gate*) and show that below that gate it is a regularizer wearing a discovery costume. Using a campaign of 79 controlled experiments spanning CIFAR-10/100 and a $4 \times 3 \times 6$ grid of transfer datasets (Stanford Cars, Flowers-102, DTD, Galaxy10 DECals) $\times$ pretraining bases (DINO, LeJEPA, VISReg) $\times$ contrastive/regularized objectives, we establish four further results. (i) Detection and calibration are *empirically dissociated*: softmax-contrastive spaces win the probe and score 0.000 per-event detection power at their conventional temperature, while density-ratio-calibrated spaces invert the ordering; no CIFAR-100 space in our sweep exceeds probe 0.94 with tied-Mahalanobis $\mathrm{AUC} > 0.42$. (ii) The dissociation is *repairable by construction rather than by loss design*: residual fine-tuning, which trains a second head on $\mathbf{z} - \boldsymbol{\mu}_y$ and concatenates, beats the alternative (iterated anchor discovery) in $12/12$ dataset$\times$base cells and yields the first spaces that hold a record probe and calibrated distances simultaneously. (iii) *Scale-free* statistics see novelty that distance-based scores cannot: local intrinsic dimension (LID) reaches $\mathrm{AUC} = 0.951$–0.955 on Flowers-102 using *no novelty labels*, exceeding the best supervised probe (0.906) — the only unsupervised score to beat a trained probe anywhere in the campaign. (iv) Interpretability is a *measurable, dissociable* property: superclass agreement and dendrogram purity track supervision, not probe quality, and a never-labeled held-out class reliably lands among its true semantic siblings. We distill the results into a reporting protocol and a set of regime rules — which objective, which statistic, which repair — for practitioners building embedding spaces meant to find things nobody has labeled.

1 Introduction

Representation learning is evaluated, overwhelmingly, by how well a linear classifier reads a representation out. This is a defensible proxy when the downstream task is classification into known categories. It is the wrong proxy when the downstream task is *discovery*.

A discovery pipeline in the physical sciences — searching for a resonance in collider data, an anomalous transient in a survey, an unfamiliar defect in a microscopy campaign — asks a different question of an embedding. It asks: *given a reference population, is this event unlike anything I have seen, and by how much?* Answering requires the space to support an absolute notion of distance, so that a threshold set on simulated background transfers to data. It requires the tail of the distance distribution to be enriched in genuine novelty, so that clustering the tail recovers new categories rather than mislabeled old ones. And — because a discovery must eventually be *explained* — it requires the geometry to be semantically legible, so a domain expert can say what the new cluster sits next to and why.

None of these are linear separability. All of them are measurable.

The LeJEPA opening The recent line of work that regularizes embeddings toward an explicit distributional target — most directly LeJEPA and its sketched isotropic Gaussian regularizer (SIGReg), which penalizes the sliced 1-D Wasserstein-2 distance between the embedding and $\mathcal{N}(0, I)$ — changes what is available to measure. Once a marginal is *specified* rather than emergent, distances have units: a Mahalanobis radius means something, a density ratio can be calibrated, and a likelihood-based anomaly score becomes well posed. Our starting hypothesis is that this is the correct substrate for discovery, and that its evaluation should exploit the structure it provides instead of discarding it in favor of a probe.

That hypothesis turns out to be half right, in an informative way. A Gaussian marginal is necessary for calibrated geometry but not sufficient for detection, and the two properties trade off sharply enough that we treat them as separate axes throughout.

Contributions

- **A four-currency evaluation battery** (§3) for embedding spaces intended for discovery: detection, calibration, discoverability, interpretability — with concrete estimators, standard protocols, blind spots, and known failure modes for each.
- **Systematic evidence that the currencies dissociate** (§4), across 79 experiments, two image regimes, four transfer datasets, three pretraining bases, and ten training objectives. The dissociation is not noise and is not a tuning artifact: it is reproduced under paired seeds and inverts predictably with the number of seen classes.
- **A unified account of new-class discovery** (§5), which we regard as the central contribution. The Neyman–Pearson functional minimized during representation learning is, verbatim, the SparKer detection statistic evaluated at event rather than pair level; and the SIGReg marginal that licenses the first is what makes the clustering step — Euclidean k-means with a unit-variance BIC — correctly specified rather than merely convenient. We give the *purity gate* that determines when the pipeline finds real classes, show that below it the loop is a regularizer rather than a discovery method (established by a reversal in which the outcome flips sign while purity does not change), and document a bandwidth pathology in kernel two-sample testing that we expect to affect comparisons beyond this work.
- **A methodological result we did not set out to find** (§9): across five independent interventions, *repairing a theoretical identity did not improve the space*. Each repair was well-motivated, each succeeded on its own terms, and none delivered the downstream benefit its motivation implied. We think this generalizes beyond this program and state it as a caution for principled-objective design.
- **Two repairs, benchmarked head to head** (§6): iterated anchor *discovery* and *residual fine-tuning*. Residual fine-tuning wins $12/12$ cells and produces the campaign's records.
- **Local intrinsic dimension as a first-class novelty statistic** (§7), including a full control study: holdout rotation, distance baselines, bootstrap intervals, per-class decomposition, and k-sensitivity. It sets a new unsupervised record on Flowers-102 and, importantly, we characterize where it *fails*.

dataset	classes	held out	seen train pool	character
CIFAR-10	10	$\{4\}$	45 000	coarse, many samples/class
CIFAR-100	100	$\{4\}$	49 500	many-class, low resolution
Stanford Cars	196	186–195	7 712	fine-grained (car models)
Flowers-102	102	92–101	1 840	fine-grained, few-shot ($\sim$18/class)
DTD	47	37–46	2 960	texture
Galaxy10 DECals	10	$\{9\}$	1 585	coarse morphology, 10%/90% split
FGVC-Aircraft	100	90–99	$\sim$6 600	fine-grained (variants)

Table 1: Datasets and open-world splits. Transfer bases are DINO, LeJEPA and VISReg ViT-B/16 trunks (768-D CLS features); CIFAR uses a ResNet-20 trunk (CIFAR-pretrained, or randomly initialized for the from-scratch lineages).

- **Interpretability as a metric, not a figure** (§8): superclass agreement@k, dendrogram purity, centroid silhouette, and cross-space similarity (CKA, mutual-kNN, Procrustes, ridge R^2, TwoNN intrinsic dimension), applied to every space in the campaign.

2 Setup

Open-world protocol In every experiment a set of classes is held out *entirely*: their images and their labels are removed from the training corpus, so at evaluation time they are genuinely novel. Unless noted, the held-out set is the final ten classes of the dataset (Galaxy10: the final one; CIFAR: class 4). Embeddings are then frozen and scored. This is stricter than the common "train-on-everything, test-novelty-by-relabeling" protocol, and we flag the one legacy table (§4.2, aircraft trunk reuse) where an earlier run violated it.

Objectives All objectives are instances of one design cube, which lets us attribute effects to axes rather than to named methods. An objective is specified by four coordinates:

$$\text{(positives)} \times \text{(critic)} \times \text{(estimator)} \times \text{(marginal)}.$$

Positives are instance (augmentation) pairs or supervised (same-label) pairs. The *critic* g is cosine $\langle \hat{\mathbf{z}}, \hat{\mathbf{z}}' \rangle/\tau$, bilinear $\langle \mathbf{z}, \mathbf{z}' \rangle/\tau$, or distance $-\frac{1}{2}\|\mathbf{z} - \mathbf{z}'\|^2/\tau$. The *estimator* is either the standard row-normalized softmax (recovering NT-Xent/SupCon) or the un-normalized maximum-likelihood form

$$\mathcal{L}_{\text{NPLM}} \;=\; \mathbb{E}_{\text{ref}}[e^{g} - 1] \;-\; \mathbb{E}_{\text{pos}}[g]\,, \tag{1}$$

whose minimizer is the *absolute* log density ratio $g^\star = \log \frac{p(\mathbf{x},\mathbf{x}')}{p(\mathbf{x})p(\mathbf{x}')}$ with the calibration constraint $\mathbb{E}_{\text{ref}}[e^{g^\star}] = 1$. The *marginal* is none, global SIGReg, or classwise SIGReg. The distinction in the estimator axis is the crux of this paper: softmax self-normalizes per row and therefore recovers pointwise mutual information only up to an unknown per-row constant, destroying absolute scale; Eq. 1 does not. We verify the calibration empirically against analytic PMI on a controlled toy ($r > 0.9$).

Two knobs, one inert Two hyperparameters govern the NPLM arms, and they are not symmetric. At initialization the interaction term of Eq. 1 is $\sim 10^{10}$ while $\lambda \cdot \text{SIGReg}$ is $\sim 10^{-2}$: λ *is inert for the first $\sim$2 epochs* and only shapes the endgame. By contrast τ is a sharp optimum ($\tau = 1$ for bilinear on CIFAR-100; $\tau \in \{0.5, 2\}$ each cost 0.06–0.10 probe AUC). We therefore scan τ and report λ ablations only where the marginal type changes.

3 A four-currency battery for discovery

We propose that any embedding space advertised for discovery be reported in four currencies. The first two are per-space properties; the third is a property of a *pipeline*; the fourth is a property of the space's relation to human categories.

arm	positives	critic	estimator	marginal	labels
`simclr`	instance	cosine	softmax	—	no
`lejepa`	MSE invariance $+\lambda$·SIGReg			global	no
`ss` (`simclr_sigreg`)	instance	cosine	softmax	global	no
`supcon`	supervised	cosine	softmax	—	yes
`supcon_sigreg`	supervised	cosine	softmax	global	yes
`nplm_bilinear`	instance	bilinear	NPLM	global	no
`nplm_distance`	instance	distance	NPLM	global	no
`nplm_sup_dist`	supervised	distance	NPLM	global	yes
`nplm_*_cw`	(either)	(either)	NPLM	classwise	varies
`sup` (recipe)	prototype CE to anchors + repulsion			classwise	yes

Table 2: The objectives as coordinates in the design cube. `lejepa` and the `sup` recipe are not contrastive and are listed by their components. Standard training: 20 epochs, Adam 10^{-3}, batch 256 two-view (balanced 24/class for classwise arms), $\lambda = 1$.

3.1 Currency I — Detection

Can novelty be read out at all? Estimator: the AUC of a linear holdout-vs-rest probe trained on frozen embeddings, reported as mean±sd over 3 seeds. This is the standard metric and we retain it, with two caveats that recur throughout: single-seed spread is ≈ 0.017, so **gaps below** 0.02 **require 3–5 seeds**; and the probe is blind to radial and non-linear novelty structure, which is precisely the structure the other currencies measure.

We also report nearest-centroid top-1 accuracy over seen classes (`acc`) and the macro one-vs-rest AUC of the prototype posterior (`supAUC`) as closed-set sanity checks — a space can win the novelty probe while being useless at its nominal task, and we have observed exactly that (`nplm-bil-ft` on Flowers: probe 0.808 post-discovery, `acc` 0.531).

3.2 Currency II — Calibration

Do distances have absolute meaning? A discovery threshold is set on a reference population and applied to data; if the score's scale drifts, the threshold is worthless even when the space is geometrically fine. We estimate calibration at two levels.

Per-space: novelty AUC of the minimum distance to a seen-class centroid (`eucl`), and of tied and per-class Mahalanobis distances (`mahaT`, `mahaPC`; shrinkage 0.1). Plus a Gaussianity summary — eigenvalue conditioning, class RMS, maximum off-diagonal correlation, sliced-Wasserstein ratio, skew and kurtosis — which audits whether the SIGReg target was actually reached rather than merely optimized.

Per-event: the fraction of injected signal events exceeding the $\alpha = 0.05$ null quantile of the reference score distribution. This is the metric that most sharply separates the objective families, and it is the one closest to how a physics search actually operates. **SupCon-family spaces score** 0.000 **here** at the conventional temperature $\tau = 0.1$, on every dataset and at every signal fraction. We first read this as a property of self-normalized objectives as such. It is not: it is a property of that temperature. Raising τ to 1 *with* a SIGReg marginal yields per-event 0.04–0.06 (§4.1). The correct statement is the one App. A actually proves — the softmax *interaction* cannot supply absolute scale — and whether the *space* has usable scale then depends on whether the marginal is allowed to supply it instead, which a sharp temperature prevents.

Dataset-level: three two-sample batteries — SparKer (semi-parametric kernel discrepancy, M=16 kernels), Mahalanobis, and MMD — run as toy experiments (N_D events per toy, signal fraction f injected from the holdout pool, power = fraction of 50 signal toys above the $\alpha = 0.05$ quantile of 200 null toys). Two lessons are protocol-critical. First, **kernel statistics must use an annealed median-heuristic bandwidth**: a fixed σ=1 is calibrated only for the unit-class-width supervised recipe, and a discovery fine-tune that rescales a compact trunk by $\sim 4\times$ sends fixed-σ SparKer to zero power at every fraction *while the space is fine* (annealed: 1.00). We regard this as a general warning about kernel two-sample tests across heterogeneous geometries. Second, statistic and

space (CIFAR-10, 32-D, holdout 4)	probe	acc	`eucl`	`mahaT`	per-event
`supcon`	**0.942**	0.861	0.689	0.644	0.000
`supcon_sigreg`	0.882	0.844	0.637	0.577	0.000
`sup` (supervised SIGReg)	0.923	0.884	0.778	0.645	—
`nplm_sup_dist`	0.783	0.871	0.726	**0.723**	**0.20**
`nplm_bilinear` (no labels)	0.853	0.546	0.510	0.494	—
`simclr`	0.867	0.729	0.236	0.229	0.000
`lejepa`	0.864	0.695	0.408	0.371	—

Table 3: The dissociation in one table, at the conventional $\tau = 0.1$; §4.1 shows the per-event column is temperature-dependent. The probe ordering and the calibration ordering are close to reversed. Note `simclr`'s `eucl` of 0.236 — *below* chance-adjusted expectations for a space with a probe of 0.867: the information is there and is not in the distances.

geometry must be matched: parametric statistics need calibrated spaces, and Mahalanobis is simply dead on CIFAR-100 at every dimension and every space (inter-class similarity), where MMD remains the dimension-robust choice.

3.3 Currency III — Discoverability

Does the tail contain the new thing? A discovery loop pools events past a high quantile (0.95) of the seen-distance distribution, selects a cluster count by BIC, merges nearby anchors, and fine-tunes the space on seen labels plus the resulting pseudo-labels. The diagnostic that matters is **pool purity**: the fraction of the pooled tail that is genuinely novel. Everything else about the pipeline is downstream of it.

Purity varies over two orders of magnitude across our datasets — 0.795 on DTD, 0.33–0.66 on Flowers, 0.34 on CIFAR-10, 0.03–0.14 on Cars, 0.003–0.013 on CIFAR-100 — and the variation is *structural*, not a tuning failure. On CIFAR-100 the pool is dominated by background outliers rather than by novelty, so the loop is blocked *geometrically* (§5.10) — we initially misread this as a rate limit. On Cars, purity caps at 0.14 for a deeper reason we consider one of the campaign's cleanest negative results: **a novel car model looks like existing car models**. Pooling requires novelty that is geometrically *outlying*, and fine-grained novelty is not, calibrated space or no. We recommend reporting purity alongside any claimed discovery gain, because a gain at low purity is mechanistically something else (§6).

3.4 Currency IV — Interpretability

Is the geometry legible? Where a superclass partition exists (CIFAR-100 coarse labels, aircraft manufacturer, car make, galaxy morphology) we compute: nearest-class-centroid superclass agreement@1 and @5 against chance, average-linkage dendrogram purity, centroid silhouette, and the within/between distance ratio. Where it does not, we report top-5 nearest-centroid tables, which are directly auditable by a domain expert.

We additionally characterize how spaces relate to *each other*, on aligned samples (n=4000/cell): linear CKA, mutual kNN@10, LLE-weight transfer, orthogonal Procrustes, directional ridge-map R^2, calibration transfer through a fitted ridge map, and TwoNN intrinsic dimension. This turns "these two objectives learn different things" from an assertion into a measurement, and it is how we establish that the NPLM spaces are *reorganizations* of information the softmax spaces already contain rather than new information (§8).

4 The central dissociation

The single most robust finding across the campaign is that Currency I and Currency II are *anti-correlated across spaces*. Softmax-contrastive objectives maximize linearly readable structure and carry no absolute distance information; density-ratio-calibrated objectives are ready-to-use metric spaces at a probe cost.

On CIFAR-100 the effect was sharp enough that we stated it as a constraint: **no space in the**

sweep exceeds probe 0.94 **with** `mahaT` > 0.42, across 30+ standalone spaces at 32/64/100-D, hub-initialized and from-scratch, with the probe ceiling (≈ 0.95) owned by supervised softmax and prototype recipes and the `mahaT` ceiling (≈ 0.47–0.49) owned by NPLM arms.

The calibration ceiling is not a ceiling That second half has since been broken, and by the arm we least expected. In a controlled dimension sweep whose *control* was plain `supcon_sigreg`, the control posts `mahaT` $= 0.545$–0.558 at 100–128-D over three seeds (± 0.03) while holding probe 0.90–0.91 — comfortably above the 0.47–0.49 band that every NPLM arm had established as the CIFAR-100 limit. A *softmax* objective with a Gaussian marginal, given enough width, is the best single-loss both-currencies CIFAR-100 space on record.

This does not overturn the dissociation as stated — probe 0.91 is below the 0.94 threshold in the constraint, so the specific claim survives — but it substantially weakens the reading we had attached to it. "Calibrated geometry belongs to the density-ratio family" is false as a general statement; what is true is that the density-ratio family reaches calibrated geometry *at low dimension and without labels*, which is a narrower and more contingent claim. We flag this as the clearest instance in the campaign of a ceiling that was an artifact of which arms had been swept rather than a property of the problem, and we would treat the remaining ceilings in this paper with corresponding suspicion. That suspicion was warranted: a second one fell within days (§5.10).

4.1 The cause: a temperature × marginal interaction

The ceiling-breaking arm was found by accident, and tracing it is worth the space because the lesson is methodological.

The configuration that broke the ceiling never set its temperature, and so ran at the loss class's default $\tau = 1$ rather than the canonical SupCon value 0.1. Under a controlled factorial the effect is unambiguous, and it is an *interaction*:

arm (τ, marginal)	32-D	64-D	100-D	128-D	200-D
`supcon` (0.1, none)	0.356	0.342	0.344	0.363	0.368
`supcon_sigreg` (0.1, on)	0.404	0.367	0.382	0.399	0.354
`supcon` (1.0, none)	0.410	0.395	0.425	0.420	0.395
`supcon_sigreg` (1.0, on)	**0.539**	**0.557**	**0.562**	**0.572**	0.492

Table 4: CIFAR-100 tied-Mahalanobis novelty AUC, 3 paired seeds. Only $\tau = 1$ *and* the marginal together clear the 0.47–0.49 band; each alone stays under it.

Neither knob alone suffices, which is why every one-knob scan in the campaign missed it. The effect is fully present at 32-D, so the dimension we originally credited is incidental. The resulting space — mahaT 0.572 ± 0.031 at 128-D with probe 0.910 — is the best single-loss both-currencies CIFAR-100 space on record, and is now designed rather than accidental. An independent intervention (§8.2) reaches numerically the same place, so this is a basin rather than a knife-edge.

The mechanistic reading connects to App. A. A sharp temperature makes the softmax interaction dominate, and the interaction provably cannot supply absolute scale. Loosening it lets the SIGReg marginal shape the full covariance instead — which is where the calibration comes from. Consistently, the winner sits at RMS ≈ 0.50 while the $\tau = 1$ arm *without* the marginal reaches RMS ≈ 1.0 with poor mahaT: unit width is neither necessary nor sufficient, and what $\tau = 1$ buys is a looser interaction, not a better width.

How far the basin extends A follow-up scan settles the scope, and the answer limits the damage. Sweeping $\tau \in \{0.05, \ldots, 3.0\}$ against the marginal on both CIFAR datasets, the basin is **many-class only**. On CIFAR-100 it is a *plateau* from $\tau \geq 0.3$ rather than a knife-edge at 1.0 (mahaT $0.238 \to 0.506$, probe holding 0.917–0.926, and these are the only cells with nonzero per-event). On CIFAR-10 the same move buys 0.03 of mahaT — noise-scale — and collapses the probe from 0.94 to 0.64–0.67. So $\tau = 0.1$ is correctly tuned for few-class data and under-tuned only for many-class data, which is the regime we have already re-run. No campaign-wide re-tuning follows. We

regime	example	probe champion	calibration champion
few classes, coarse	CIFAR-10, Galaxy10	`supcon`	`nplm_sup_dist`
many classes, fine-grained	Cars, Flowers, Aircraft	`ss/supcon_sigreg`	`ss`
texture	DTD	`simclr`	`ss`
many classes, low-res	CIFAR-100	`supcon`	`nplm_bilinear`

Table 5: Regime rules, replicated on all three pretraining bases. Note the inversions: the SIGReg marginal *helps* on fine-grained low-data transfer and *hurts* on CIFAR; label-free instance discrimination wins outright on texture, where labels add nothing to the probe.

note one unresolved discrepancy: the scan gives mahaT 0.502 at 100-D where the dedicated factorial gave 0.562 ± 0.037, about 1.5 combined standard deviations apart.

The uncomfortable part $\tau = 0.1$ was inherited as "the SupCon temperature" and held fixed across the entire campaign; it was never a measured choice. The best CIFAR-100 space we have was therefore found by a configuration slip, and an entire continuous axis under the family that wins most of our probe comparisons had gone unswept. We report this plainly because it bounds how much weight the comparative claims in this paper can carry: they compare arms *at their inherited defaults*, which is not the same as comparing arms at their best.

The frontier between the two currencies is otherwise populated only by *concatenations* (§6).

This is not a seed artifact The one place we initially believed the dissociation was breakable — `nplm_bilinear` reaching probe 0.935 on CIFAR-100, apparent parity with `supcon` — dissolved under paired multi-seed testing: over 5 paired seeds `supcon` gives 0.940 ± 0.004 against `nplm_bilinear`'s 0.855 ± 0.042, paired difference -0.085 ($t = -4.7$). The single-seed 0.935 was a tail draw from a heavy-tailed seed distribution. The $10\times$ larger seed variance of the label-free NPLM arm is itself an unexplained and, we think, interesting phenomenon.

The dissociation recurs at every level It is not a property of the loss family alone. It reappears (i) in the fine-tuning objective: prototype/repulsion fine-tuning wins the probe (0.97 vs 0.81–0.84) while NPLM+SIGReg fine-tuning wins *every* power statistic (per-event 0.57 vs 0.30–0.48; SparKer $1.00/0.92$; Mahalanobis 0.90; MMD 0.74) and holds round-2 pool purity (0.22–0.30 vs 0.03–0.11); (ii) in the residual construction (§6), where NPLM residuals preserve calibration and plain residuals buy probe with it; and (iii) in semantic structure (§8), where the calibration-strong label-free arms are also the semantically emptiest. The practical rule is uncomfortable but clear: **choose by consumer**. Linear readout $\rightarrow$ prototype/softmax. Detection with a transferable threshold $\rightarrow$ NPLM.

We qualify this in §5.8, where it turns out that a substantial part of what we are calling a dissociation is a property of the fine-tuning *update* rather than of the space: on fine-grained data, omitting the update recovers both currencies at once. The dissociation between *standalone objectives*, which is what Table 3 measures, stands.

4.2 Regime rules: which corner wins where

The winning objective is determined by regime, and — importantly for practitioners — is *base-invariant*: the same arm wins on DINO, LeJEPA and VISReg trunks. The governing variable is the number of seen classes relative to the embedding dimension, modulated by samples/class.

Three sub-rules are worth isolating.

The classwise marginal is gated by optimization, not by geometry. We previously stated this rule geometrically: classwise SIGReg should beat global SIGReg *iff* the anchors are reachable, $\dim \geq n_{\text{classes}}$, since with 100 anchors in 32-D the centroids stall ≈ 3.5 from their targets. That rule is wrong, and the experiment that tested it is worth reporting in full because the failure is informative. Running CIFAR-100 classwise arms at 100-D, 128-D and 200-D — all geometrically realizable, the last comfortably so — the centroid-to-anchor distance **never collapses**: 3.46, 3.47, 3.51, indistinguishable from the 32-D stall, across all nine runs. The anchors are unreachable for an *optimization* reason: the loss equilibrium parks centroids about 3.5 from their targets regardless of whether the geometry permits reaching them. Nor does the extra dimension help the metric it was supposed to help — classwise mahaT *declines* with dimension ($0.468 \to 0.326$), so the encouraging 100-D value in our archive was a fair draw rather than a door. Dimension is neither the enabler nor the fix. We still recommend reporting centroid-to-anchor as a routine diagnostic, but as a diagnostic of the *optimization*, and a practitioner should not expect to buy realizability with width.

High dimension is not free for calibrated objectives. Moving CIFAR-100 from 32-D to 64-D causes the bilinear NPLM arms to *regress on both currencies* ($0.935 \to 0.858$; $0.925 \to 0.825$), while softmax and supervised distance arms gain mildly. On Cars at 100-D the same signature appears from the other side: `nplm_sup_dist` gains the most probe ($+0.096$) but loses its per-event calibration lead ($0.106 \to 0.03$). At high class counts the extra dimensions are spent on probe-readable directions, not calibrated ones.

Fine-tuning inverts the base ranking. On frozen features DINO dominates every currency. After end-to-end fine-tuning the ordering largely dissolves — VISReg holds several pre-discovery records (DTD `simclr-ft` 0.854, Cars `ss-ft` 0.741). Cross-space CKA supplies the mechanism (§8): supervision pulls every base toward one shared geometry.

A two-stage recipe, now citable. Fine-tuning a trunk with supervised NPLM and then fitting a `supcon_sigreg` head was the campaign's most promising *uncitable* result, carried with a holdout-contamination caveat. Re-run under the strict open-world protocol the numbers move by only 0.007–0.016 ($0.787/0.796/0.805$ probe on DINO/LeJEPA/VISReg, with mahaT up to 0.814), so the caveat can be struck. As predicted it lands between the single-stage parents and the residual champions (0.816–0.866), making it a real recipe that the residual construction nonetheless beats.

Cross-entropy is not the ceiling. On every substrate tested, a `supcon_sigreg` head beats a supervised CE head on every column, including — on a LeJEPA fine-tuned trunk — beating CE's own trained-classifier top-1 with mere nearest-centroid accuracy (0.638 vs 0.625).

5 Discovering new classes: one likelihood model, three scales

We now come to what we regard as the central result of this work. The preceding sections evaluated embedding spaces; this one uses them, and the finding is that the pieces of an open-world discovery pipeline — the representation objective, the clustering that proposes new classes, and the statistical test that certifies them — are not three independent design choices. Under a specified Gaussian marginal they collapse into **one likelihood model instantiated at three scales**, and the places where the pipeline fails are exactly the places where that identification breaks.

5.1 The same Neyman–Pearson functional, twice

The representation objective of §3, written over pairs, is

$$\mathcal{L}_{\mathrm{NPLM}}[g] = \mathbb{E}_{p(x)p(x')}\big[e^{g} - 1\big] - \mathbb{E}_{p(x,x')}\big[g\big] \qquad \text{(Eq. 1, repeated).} \tag{2}$$

The dataset-level detection statistic we use, SparKer (Grosso et al., `arXiv:2511.03095`), trains a kernel model f on a data sample D against an anomaly-free reference R by minimizing

$$\mathcal{L}_{\mathrm{NP}}[f] = \sum_{x \in R} w\big(e^{f(x)} - 1\big) - \sum_{x \in D} f(x), \qquad w = N_D/N_R, \tag{3}$$

and reports $t_{\mathrm{NP}} = -2\,\mathcal{L}_{\mathrm{NP}}$, an asymptotic log-likelihood-ratio (Wilks) statistic.

These are the same functional. Eq. (3) is Eq. (1) with pairs replaced by events and the product-of-marginals reference replaced by a control sample. The consequence is not cosmetic: **the objective that shapes the embedding is the objective that later tests it**, so the quantity being estimated during representation learning and the quantity being thresholded at detection time are the same calibrated log density ratio.

What the unification does *not* give you We tested the strongest reading of this claim and it failed, so we state the limit before developing the consequences. If the two scales were the same object learned twice, the representation-level critic should transfer to the event-level test: initializing SparKer's kernel centres at the trained seen-class centroids ought to save it the work of rediscovering them. It does not. Warm-starting is neutral to *actively harmful* — at a 50-step budget on CIFAR-10, power at f=0.02 falls from 0.800 (cold, data-sample init) to 0.560 (warm) for `nplm_sup_dist` — and the full-budget t_{NP} trajectories are statistically indistinguishable. The mechanism is instructive: class centroids sit exactly where the corpus-to-reference density ratio is ≈ 1, so warm-started centres must first migrate *away* from their initialization, while a data-sample initialization occasionally seeds a centre directly on a novel event. **The two scales share the functional form and the loss, not transferable learned content — the representation's class structure is precisely where the novelty is not.**

A second experiment closes off the stronger version from the other direction. If the event-level statistic is the thing we care about, the natural move is to optimize the representation *through* it: alternate between fitting the kernel model and updating the encoder to raise t_{NP}. Used as the discovery fine-tuning objective on CIFAR-10, this is worse than both existing alternatives on *every* column — including the event-level statistic it directly optimizes, where post-discovery SparKer power collapses to ≈ 0.1 against 1.00 for prototype and NPLM fine-tunes. The mechanism is statistic-chasing: pseudo-novel points climb the *current* f surface, class structure smears (only the global SIGReg term still restrains it), and the refitted kernel then finds less genuine separation than before. Optimizing a representation through a fitted density-ratio model destroys the separation that model exists to measure.

Together these bound the unification empirically and in both directions. **The Neyman–Pearson loss transfers across scales as a *form* — it licenses the estimand, the calibration, and the clustering (§5.2) — but neither its learned parameters nor its optimization transfer.** We think this is the honest statement of what the identification buys, and we would discourage the obvious end-to-end variant on this evidence rather than on taste. The $e^f - 1$ term plays an identical role in both — it pins $\mathbb{E}_{\mathrm{ref}}[e^f] = 1$ and thereby fixes the additive constant — which is why a softmax-trained space, whose scale is a gauge choice (App. A), has nothing for a Neyman–Pearson test to threshold and scores 0.000 per-event power at $\tau = 0.1$ — though see §4.1, where a looser temperature lets the marginal supply the scale the interaction cannot.

5.2 Why clustering is correctly specified in a calibrated latent

The third scale is the clustering step, and the same marginal constraint makes it well posed. Our discovery loop selects the number of new classes by BIC under *unit-variance* isotropic Gaussians, i.e. with per-component log likelihood $-\frac{1}{2}\|z - \mu_k\|^2 + \text{const}$ and penalty $k\,d \log n$. That model is ordinarily a crude approximation. Under SIGReg it is not: once $p(z) = \mathcal{N}(0, I)$ is enforced, Euclidean distance *is* Mahalanobis distance, so k-means is the correct estimator for this likelihood and the BIC is evaluated under the same density the representation was trained to have. The clustering, the detection statistic, and the loss are all reading the same quadratic surface (§A, final subsection).

This is the constructive payoff of the LeJEPA program that motivated the work. Specifying the marginal is not only a regularizer that improves numbers; it is what licenses the downstream inference machinery to be applied without misspecification.

Where the specification is only partial, and what it costs Two honest caveats. SIGReg constrains the *aggregate* marginal, not the per-class conditionals, so "Euclidean = Mahalanobis" holds globally but not necessarily within a discovered cluster; the classwise marginal repairs this exactly when anchors are realizable (§4.2), which is a checkable precondition and frequently fails. And the unit-variance BIC has a documented pathology at calibrated scale: for a cluster of unit width the $k\,d \log n$ penalty is smaller than the sum-of-squares gain from splitting, so BIC saturates at $k_{\max}$ and *fragments* a single true class. We do not regard this as a bug to be tuned away — it is the correct behaviour of that likelihood — and the loop absorbs it downstream with a weighted union-find merge of anchors closer than a threshold, plus a repulsion exemption between discovered anchors. Fragmentation is nonetheless the mechanism behind the worst outcome we record (`ss-ft`/LeJEPA on Flowers, §5.7).

5.3 The pipeline

Concretely, one discovery round is: (i) embed the unlabeled corpus and score every point by its distance to the nearest seen-class anchor; (ii) *pool* the points past the 0.95 quantile of the seen-population score; (iii) choose k by BIC and cluster the pool; (iv) merge anchors closer than

Aircraft, frozen heads, SpK@f=0.1				**Cars**, e2e ft, SpK@f=0.1	
arm	DINO	LeJEPA	VISReg	arm	DINO
`nplm_sup_dist`	**0.64**	0.28	0.68	`ss-ft`	**1.00**
`supcon`	0.42	**0.64**	**0.90**	`supcon-ft`	0.98
`supcon_sigreg`	0.38	0.52	0.36	`nplm-sup-ft`	0.94
`simclr_sigreg`	0.18	0.12	0.10	`simclr-ft`	0.30
`nplm_bilinear`	0.16	0.18	0.08	`nplm-bil-ft`	0.12
`lejepa`	0.16	0.04	0.06	`sigreg-ssl-ft`	0.12
`nplm_distance`	0.14	0.12	0.10		
`simclr`	0.06	0.44	0.22		

Table 6: SparKer detection power at $\alpha = 0.05$. Aircraft columns use a fixed $\sigma = 1$; all spaces there share the compact frozen-head scale, so comparison *within* the table is fair but not against other tables — the bandwidth caveat above, applied to our own numbers. Cars columns use the annealed schedule.

`merge_dist`; (v) assign pseudo-labels and fine-tune the space on seen labels plus pseudo-labels with a prototype/repulsion objective, co-learning the new anchors. Two rounds of five epochs is the settled configuration. The diagnostics recorded per round are the pool size, the **purity** (fraction of the pool that is genuinely novel), $\hat{k}$, the anchor count, the margin AUC of $d_{\text{seen}} - d_{\text{disc}}$, and the per-holdout anchor AUC.

5.4 SparKer: what it measures and how it must be run

SparKer models the log density ratio as a sparse self-organizing kernel mixture, $f(x) = a^\top [\, p_{[k]}(x) \cdot k(x)\,]$ with Gaussian kernels $k_i(x) = \exp(-\|x - \mu_i\|^2/2\sigma^2)$ and a gating $p_i = k_i / \sum_j k_j$; both the M kernel centres μ_i (initialized from the data sample) and the weights a are trained on Eq. (3). We use $M = 16$ centres and 300 steps. The bandwidth is *annealed* linearly from σ_0 to $\sigma_0/10$, with t_{NP} recorded at three checkpoints, so a single run probes several resolutions; per-σ p-values are calibrated on anomaly-free toys and aggregated as $-\frac{1}{2}\min_\sigma \log p_\sigma - \frac{1}{2}\,\overline{\log p_\sigma}$, with detection declared above the $(1-\alpha)$ quantile of the aggregate's own null. Power is reported with Clopper–Pearson intervals over 50 signal toys against 200 (pre) or 100 (post) null toys at $\alpha = 0.05$.

The bandwidth rule, and why it generalizes beyond us The single most costly protocol error in the campaign was running SparKer at a fixed $\sigma = 1$. That value is calibrated for the unit class width of the supervised recipe and *only* for it. A discovery fine-tune rescales a compact NPLM trunk by roughly $4\times$; at fixed σ the test then reports zero power at every signal fraction *while the space is perfectly fine* — the annealed schedule scores 1.00 on the same embeddings. We therefore default to the annealed median-heuristic schedule for any comparison across spaces whose geometry may differ, and we flag the affected historical rows rather than silently restating them. The general statement is that a kernel two-sample test carries an implicit length scale, and comparing such tests across representations *silently compares length scales unless the bandwidth is set adaptively*. We suspect this affects published comparisons beyond this work.

5.5 SparKer results

Table 6 gives the picture. **SparKer power is supervised-dominated**: on Cars the three label-bearing arms reach 0.94–1.00 while every label-free arm sits at 0.12–0.30, an eight-fold separation far larger than the corresponding probe gap (0.73 vs 0.59–0.62). Kernel detection is the currency in which supervision matters most, which is the mirror image of DTD, where labels add nothing to the probe.

Two finer points. First, the champion is base-dependent for SparKer (`nplm_sup_dist` on DINO, `supcon` on LeJEPA and VISReg) even though it is *not* base-dependent for MMD, where `nplm_sup_dist` leads on all three (0.86/0.72/0.78). The two kernel statistics disagree about which space is best while agreeing that supervision wins — statistic–geometry matching operating at a finer grain than we can currently explain, and a caution against reporting a single kernel test. Second, on CIFAR-10 SparKer is the statistic that most clearly rewards the discovery step under the corrected bandwidth:

the supervised-NPLM classwise space moves $0.42 \to 0.66$ under an NPLM fine-tune, while the prototype fine-tune that wins the probe moves it $0.42 \to 0.26$ — the probe/power dissociation of §4, visible in the kernel test.

5.6 The residual child is a detector, not just a probe booster

An earlier draft of this paper reported the two tables above and noted that the residual and concatenated spaces of §6 — which hold every dataset record here — had *no* SparKer numbers, because the aircraft end-to-end cells were run with power disabled and the residual grid never ran batteries. We have since closed that gap: 106 previously uncovered spaces across 17 cells, scored under the exp-70 protocol from cached embeddings. The result changed our reading of the residual construction.

The res-nplm residual *child* is the best dataset-level detector in the fine-grained cells, and it beats its own parent: aircraft/VISReg 0.82 at $f{=}0.05$ against the parent's 0.44; cars/VISReg 0.96 and cars/DINO 0.80 at $f{=}0.05$ for the child alone. This is a currency in which we had not previously measured it. §6 presents the residual half as the component that *retains* calibration while the parent supplies the probe; the correct statement is stronger — the calibrated residual half carries substantial detection power of its own, which the probe and `eucl` columns never revealed.

Concatenation again costs nothing: aircraft/VISReg res-cat reaches 1.00 at $f{=}0.1$ and galaxy10/VISReg res-nplm-cat 1.00 at $f{=}0.05$. Taken with §8, concatenation is now free in three currencies — probe, semantics, and dataset-level power. The base rule of §4.2 reappears here too: LeJEPA-parented residual children are systematically weak (cars/LeJEPA 0.12–0.14 at $f{=}0.1$ against VISReg's 1.00), consistent with LeJEPA parents leaving the most room for residual drift.

The child can stand alone — and stays legible Scored on its own across the grid, the res-nplm child is not merely a contributor but a *better detector than its own concatenation* on the fine-grained VISReg/DINO cells: Cars/VISReg reaches $f_{95} = 0.049$, the best transfer reach measured anywhere, against 0.063 for the concat, and Aircraft/VISReg 0.063 against 0.072. The probe cost of dropping the parent is only 0.005–0.034, far smaller than we predicted. We also predicted the child would be semantically illegible, on the strength of the plain-residual half scrambling class structure (§8); that was wrong, and the distinction matters — *plain* residuals scramble, *res-nplm* children do not, retaining superclass agree@1 of 0.43–0.53 against 0.04 chance. A detection-only pipeline on these cells can therefore halve the embedding, improve reach, and still explain what it found. The falsifier fires on DTD and Flowers, where standalone children never cross 0.95 while their concats bracket ≈ 0.08: there the concatenation is doing the work.

The split is deployable Carried through to a full pipeline on the three qualifying cells — detect with the child, consult the parent only to explain what was flagged — child-only detection matches the concatenation within 0.005–0.007 on Cars and *beats* it on Aircraft/VISReg ($+0.048$ per-event), with better reach on all three, at half the embedding width per event. Explanations are preserved: parent-only nearest-centroid tables overlap the concatenation's at 0.73–0.85 top-1, and the parent is always closer to the concatenation than the child is. Removing the parent from the *scored* space does not change what the analyst is told.

Score regime and statistic regime coincide The most interesting entry is a negative one. **Flowers is SparKer-weak in every residual space** (best 0.94 at $f{=}0.1$, most ≤ 0.4) — the same Flowers whose LID novelty score is the strongest unsupervised result in the campaign (§7, 0.951–0.955). The on-manifold novelty that a scale-free per-event statistic detects almost perfectly is nearly invisible to a dataset-level kernel test. This is not a contradiction but a structural fact worth stating as a rule: *per-event scale-free detection and dataset-level kernel detection see different things, and the regimes are close to complementary*. A practitioner who selects a space on one of them has not selected it on the other. On CIFAR the ordering reverts to the parents (CIFAR-10 `supcon` 0.50 at $f{=}0.01$ and 0.98 at $f{=}0.02$, with res-nplm-cat the only construction close at $0.34/0.92$), and CIFAR-100 is flat below $f{=}0.05$ for every space — the injection-clamp regime of §5.7, not a property of the geometry.

Kernel count should be matched *inversely* to intrinsic dimension Having warned about the bandwidth, we scanned both of SparKer's knobs against intrinsic dimension and got one confirmation and one inversion. The annealing ratio is *flat* within toy noise across every arm and every

dataset	round-1 purity	why	discovery outcome
DTD	0.795 (`ss-ft`)	textures separate cleanly	rescues weak arms ($+0.19$)
Flowers	0.33–0.66	coarse-separable holdouts	$+0.106$ to $+0.145$
CIFAR-10	0.34	calibrated tail is the novelty	best full pipeline on record
Galaxy10	0.46 (`nplm-sup-ft`)	coarse morphology	positive 6/6 arms
Aircraft	0.32–0.53	moderate	**still probe-negative**
Cars	0.03–0.14	novel models look like seen models	probe-flat to negative
CIFAR-100	0.003–0.013	novelty is not in the distance tail	geometry-blocked

Table 7: Purity is necessary and not sufficient. The Cars and CIFAR-100 rows fail for the *same* reason once measured (§5.10); the Aircraft row fails at adequate purity, which is what forces the refined statement below.

setting: the bandwidth lesson of §5.4 concerns matching σ_0, and the anneal already covers it, so the schedule is genuinely robust in that knob. The kernel count M is not flat, and it moves opposite to our expectation. We predicted that high-dimensional spaces would need *more* kernels; in fact the high-ID `supcon-ft` space (TwoNN ID ≈ 12) does far better with *fewer* (0.60 at $M{=}4$ against 0.24 at $M{=}64$), while the low-ID `nplm-sup-ft` space (ID $\approx$ 3) mildly prefers more ($0.24 \rightarrow 0.36$). The reading is that in high dimension a larger kernel ensemble overfits the null, raising the detection threshold and costing real power — and `sigreg-ssl-ft` is killed outright at $M{=}64$ (0.00 at every ratio). Our default $M{=}16$ therefore leaves up to 0.16 of power on the table on high-ID spaces. It is tempting to promote this to a rule — match M inversely to intrinsic dimension — and we tested that by re-measuring the whole reach table with M set from each space's own TwoNN estimate. It does not survive as a recipe. Matched M helps at the *extremes* (two first-time crossings: a very high-ID coarse cell at $M{=}4$, a very low-ID texture cell at $M{=}64$) and *hurts* the mid-ID fine-grained cells. Same-M reruns put the measurement noise at 0.012–0.015, below several of the apparent differences. We therefore keep $M{=}16$ as the reach protocol and cite matched-M as a sensitivity band, under which the citable count rises from $8/17$ to $10/17$. The dataset ordering is preserved either way, so the sensitivity ranking of App. B stands.

Statistic–geometry matching No single dataset-level statistic dominates. Parametric statistics (per-event thresholds, Mahalanobis) require the calibrated spaces of §4; kernel statistics (SparKer, MMD) require scale-matched bandwidths. Mahalanobis is simply dead on CIFAR-100 at every dimension and every space, where MMD remains dimension-robust. The practical rule is to report a battery and match the statistic to the geometry, not to pick a favourite.

5.7 The purity gate: when discovery finds classes, and when it does not

Pool purity is the make-or-break diagnostic, and it varies by two orders of magnitude across our datasets for *structural* reasons.

Three conclusions, in increasing order of how much they cost us to learn.

(1) Pooling requires novelty that is geometrically outlying. On Cars, purity caps at 0.14 because a novel car model looks like existing car models: the novelty sits *among* the seen classes rather than in the distance tail, and no amount of calibration moves it there. This is the cleanest negative result in the campaign and it bounds the applicability of every tail-pooling discovery method, ours included.

(2) Above the gate, purity stops being the binding constraint. Substituting the scale-free LID score (§7) for min-anchor distance as the pool scorer triples round-2 purity — the distance threshold decays as the fine-tune inflates the space, collapsing to 0.18–0.33, while LID holds 0.60–0.68 — and *strictly dominates* as a pooling mechanism. The downstream battery does not move ($+0.000$, -0.029, $+0.007$). A mechanism can be strictly better and operationally irrelevant at the same time; only a multi-currency report reveals both facts.

(3) At low purity, "discovery" is not discovery. The sharpest evidence is a reversal. On frozen trunks the loop is probe-negative on Cars for every arm; on end-to-end fine-tuned trunks it is probe-positive for all six arms ($+0.03$ to $+0.11$) — *with purity below* 0.14 *in both cases*. Since purity did not change, the gain cannot come from pure pools. The impure-pool prototype fine-tune acts as

aircraft cell	unfrozen Δprobe	unfrozen per-evt	parent frozen Δprobe	parent frozen per-evt	**fully frozen** Δprobe	**fully frozen** per-evt
DINO	-0.037	0.330	-0.013	0.330	**0.000**	0.306
LeJEPA	-0.016	0.297	-0.020	0.213	**0.000**	0.267
VISReg	-0.012	0.526	-0.013	0.498	**0.000**	**0.523**

Table 8: Discovering anchors in a *completely frozen* space keeps 90–99% of the per-event gain at exactly zero probe cost. Freezing only the parent half — our initial hypothesis — works on DINO and fails on the other two bases.

beneficial refinement when the substrate is aligned to the task and as erosion when it is not. Reported honestly, the low-purity regime is a regularizer wearing a discovery costume, and the largest "gain" in the program is a rescue of a weak head (`nplm-sup-ft` on DTD, $0.586 \to 0.781$, replicated on all three bases), not the discovery of anything.

Finally, Aircraft refutes the tempting converse. It pools at adequate purity (0.32–0.53) and discovery *still* costs probe (-0.012 to -0.037) while buying geometry and per-event power (VISReg: per-event 0.30–0.50 across all fractions, mahaT $+0.017$). Purity is necessary, not sufficient; what fails there is the fine-tune dynamics, not the pool.

5.8 The update is the cost: frozen-space discovery

That diagnosis makes a prediction, and testing it produced the sharpest operational result in the paper. If the fine-tune *update* is what erodes the probe, then the erosion should disappear when the space is not updated — and the question becomes how much of the discovery gain survives.

Table 8 gives the answer, and it is more extreme than we predicted. Freezing only the parent half of the concatenation, which was our stated hypothesis, helps on DINO and matches the unfrozen cost on LeJEPA and VISReg. Freezing *everything* — running the pool–cluster–merge loop and adding anchors without a single gradient step on the encoder — retains 90–99% of the per-event power (0.523 against 0.526 on the record cell) at exactly zero probe and geometry cost. **On fine-grained data the per-event gain lives in the anchors, not in the space update**, and the update was almost pure cost.

Two consequences. First, round-2 pool purity stops collapsing under freezing (0.176–0.346 versus 0.023–0.065 unfrozen): a frozen space cannot inflate, so the distance threshold that decays in §5.7 remains valid. This is the LID-pooling mechanism of §7 solved from the opposite side — rather than making the scorer scale-free, remove the rescaling. Second, and more consequentially for this paper's framing: **on fine-grained cells the probe/calibration dissociation is severable**. The aircraft record probe 0.8634 is now retained *with* per-event 0.523. The "choose by consumer" rule of §4 is a statement about *update dynamics*, not an inherent property of embedding spaces, and where the update can be omitted the choice does not arise. We state the rule as the operational recipe: on fine-grained data, run discovery with the space frozen and discover anchors only.

5.9 Pooling by density ratio, in a frozen space

Two independent fixes have now appeared for the two ways the loop degrades, and they compose. §5.8 removes the cost of the update by freezing the space. The remaining question is what to *pool* on, and the NP functional answers it: rather than thresholding distance, fit SparKer on (corpus vs. seen reference) and pool on the learned density ratio — which also proposes the anchors, since its kernel centres are placed where the ratio is large.

Table 9 shows the combination. Density-ratio pooling raises round-1 purity on Cars by a third ($0.161 \to 0.214$) — the mechanism of §5.7 working as predicted, since a region can have $p_{\mathrm{data}}/p_{\mathrm{ref}} \gg 1$ at unremarkable distance — though Cars still sits below the ≈ 0.3 gate, so this is an improvement rather than a rescue. The unanticipated result is round two: SparKer pooling is *immune to the round-2 collapse in every cell.*

That immunity identifies a second failure mode we had missed. In a frozen space the distance pool *still* collapses on Cars (0.068), which rules out space inflation as the cause there; what happens

	distance pool		SparKer pool	
cell (frozen space)	purity r1	r2	purity r1	r2
cars/DINO	0.161	0.068	**0.214**	**0.215**
aircraft/VISReg	0.525	0.176	**0.591**	**0.608**
flowers/DINO	0.609	0.414	**0.618**	**0.620**

Table 9: Density-ratio pooling in a frozen space. SparKer pooling wins all six round measurements, and round-2 purity no longer decays.

question	on-manifold (Flowers, Cars, CIFAR-100)	outlying (CIFAR-10, Galaxy10)
best novelty score	LID / scale-free	distance, `eucl`
pool on	density ratio (SparKer)	distance quantile
discovery update	*freeze* the space, anchors only	fine-tune: it earns its cost
residual stacking	second decomposition helps	extra width helps
kernel test reach	poor ($f_{95} > 0.1$ on Flowers)	good ($f_{95} = 0.03$–0.07)
purity gate	blocked *geometrically*	clears

Table 10: One axis, six instruments. Each row was established independently; the column headings are the same partition every time.

instead is **anchor absorption** — the anchors discovered in round one capture novel points *below* the distance threshold, hiding them from the round-two pool. A density-ratio pool is indifferent to where the anchors are and is immune to both failure modes. Combined with the frozen space it also costs nothing: the aircraft record 0.8634 is untouched, and the anchor margin improves (0.78 vs 0.66 on Cars).

Which scorer, when Adding the NP score to the comparison of §7 completes a three-way rule, and it is not a clean sweep. On Flowers — LID's regime — the closed-form ratio statistic beats the fitted one by 0.03–0.06 in all three cells: the Neyman–Pearson lemma's optimality is asymptotic, and at these pool sizes the estimation variance of a fitted f costs more than the optimality buys. Below the gate the ordering reverses and NP is the best round-1 scorer on Cars (0.214 against 0.161 for distance and 0.133 for LID). So: *LID on on-manifold cells, NP below the gate, and either one in a frozen space*. Probes are unmoved throughout, consistent with §5.7.

A validity check that passed Because these pipelines fine-tune on the same corpus the test then examines, we checked whether the nulls remain valid. Realized false-positive rate at nominal $\alpha = 0.05$ is at or below nominal in every regime (Cars/DINO 0.040 frozen and 0.040 after a full fine-tune; Flowers conservative at 0.000 throughout), so at these sample sizes the discovery fine-tune cannot manufacture false positives and split-disjointness is not required. We flag that this clears the pipelines reported here, but not a directly adversarial objective that optimizes the test statistic itself; such a method would need its own check.

5.10 The organizing axis: is the novelty on the manifold?

The campaign has accumulated a large number of regime rules, and they turn out not to be independent. Almost all of them are the same distinction seen from different instruments: **whether the novel class sits *on* the seen data manifold or *outside* it**. We state it here because it is the single most useful thing we can hand a practitioner, and because it retires several rules that looked separate.

Three results from this round make the axis explicit rather than implied.

Frozen discovery is regime-scoped, not universal. Extending §5.8 to the remaining fourteen cells, freezing retains strong per-event power on every fine-grained cell (Flowers/VISReg 0.803 — the best Flowers dataset-level detection on record — Flowers/DINO 0.737, Cars/VISReg 0.373) but *collapses* on the coarse ones (Galaxy10/DINO 0.015, CIFAR-10 0.096 against 0.73–0.75 unfrozen). Where the novelty genuinely sits outside the manifold, the space update does real work and earns its probe cost; where the novelty is on-manifold, the update is the pure cost we identified on Aircraft.

"Choose by consumer" survives but changes axis: the choice is by novelty geometry, not by readout.

CIFAR-100 is geometry-blocked, not rate-blocked. We had attributed its 0.003–0.013 purity to arithmetic — 500 holdout images inside a $\sim$2500-event tail. A dedicated grid over holdout size and tail quantile shows otherwise: purity scales with holdout fraction but saturates at 0.121 even with a 10% novel corpus, and *stricter* quantiles make round-one purity *worse* ($0.036 \to 0.002$), because the extreme distance tail on CIFAR-100 is owned by background outliers rather than by novelty. CIFAR-100 belongs with Cars, not with CIFAR-10, and the distance pool is simply the wrong instrument there.

And the block is not structural either. Replacing the distance pool with the density-ratio pool of §5.9, in a frozen space, breaks it: round-1 purity reaches 0.358 against the distance grid's all-time ceiling of 0.121, and round-2 *rises* to 0.418 rather than collapsing. The diagnostic detail is the clean one — under the distance pool a stricter tail quantile made purity *worse*, because the extreme distance tail is owned by background outliers; under the density-ratio pool the strictest quantile is *best*. The f-ratio tail is owned by novelty. So the second of our two CIFAR-100 "structural" limits was also an artifact of the instrument, and both fell within days of each other. We now state the pattern rather than the individual results: **a limit established by sweeping one family of methods is a limit on that family, and this paper's remaining ceilings should be read that way.**

Residual stacking splits on the same line. A second residual head helps emphatically on Flowers ($+0.077$ against $+0.026$ for a width-matched control, and largest on the weakest parent seed) and not at all on Cars, where a width-matched single residual matches it — there the gain is capacity, not decomposition. On-manifold novelty rewards another decomposition step; off-manifold novelty just wants parameters.

5.11 What this means for a discovery pipeline

Assembling the above: the strongest full pipelines in the campaign are the ones where all three scales are consistent. On CIFAR-10, an NPLM-calibrated space with a supervised distance critic plus prototype discovery reaches post-hoc probe 0.96–0.98 with per-event power 0.73–0.75 against 0.59 for the supervised baseline — the calibrated tail is exactly what pooling needs, the clustering is correctly specified in that tail, and the same NP functional certifies the result. Where any one of the three breaks — rate on CIFAR-100, tail geometry on Cars, fine-tune dynamics on Aircraft — the pipeline degrades in a way the corresponding diagnostic predicts in advance. That predictability, rather than any single AUC, is what we would want carried forward.

6 Repairing the dissociation

Discovery is one of two constructions that might resolve the dissociation of §4. Having characterized it above, we benchmark it head to head against the alternative.

6.1 Discovery as a repair: substrate-dependent and purity-gated

Summarizing §5.7 in the terms of this comparison: across the full grid the discovery loop is probe-positive in $50/72$ cells, and the `supcon-ft`+discovery pipeline is positive in $12/12$ dataset$\times$base cells — the only universally safe full pipeline we found. Its gains are largest where the pre-discovery space is weakest ($+0.05$ to $+0.15$ on the unsupervised arms) and where purity is high, and it is flat or negative on fine-grained data regardless of purity.

Two failure modes bound its use as a repair. Flowers/VISReg shows that purity $0.65+$ can still leave a strong arm flat — there was nothing left to learn. And high purity does not protect against a bad *partition*: `ss-ft` discovery on the LeJEPA base collapses the probe $0.792 \to 0.613$, the worst outcome on record, by splitting round 1 into six anchors at purity 0.605 that fragment the strongest classes rather than the novelty — the BIC fragmentation of §5.2 reaching the probe before `merge_anchors` can absorb it. Treat `ss-ft`+discovery as unstable and prefer `supcon-ft` parents when a discovery step is in the pipeline.

dataset	best construction	probe (per-event)	parent	prior champion
Aircraft/VISReg	`supcon`→res-nplm cat	**0.863** (0.363)	0.809	0.812†
Cars/VISReg	`supcon`→res-nplm cat	**0.855** (0.263)	0.707	0.759
Flowers/DINO	`supcon`→res-nplm cat	**0.885** (0.334)	0.708	0.814
DTD/VISReg	`supcon`→res cat	0.847 (0.077)	0.750	0.854‡
Galaxy10/LeJEPA	`supcon`→res cat	**0.975** (0.001)	0.919	0.946
CIFAR-100	`supcon`→res-nplm cat	0.955±0.006	0.944	0.9546
CIFAR-10	`supcon`→res cat	0.9552±0.0023	0.9404	0.954

Table 11: Best residual construction per dataset (best base shown). Residual fine-tuning beats the discovery pipeline in $12/12$ transfer cells. †aircraft prior champion carries a holdout-contamination caveat; the row here uses strict open-world parents. ‡the one exp-70 record residuals do not beat (`simclr-ft` pre-discovery on texture).

6.2 Residual fine-tuning wins, and breaks the dissociation

The alternative construction: freeze the parent's seen-class centroids, train a second head end-to-end on the residual $\mathbf{r} = \mathbf{z} - \boldsymbol{\mu}_y$, and evaluate the concatenation $[\mathbf{z}; \mathbf{r}]$. The parent half keeps what supervision learned; the residual half is free to encode the within-class variation that supervision discarded — which is where fine-grained novelty lives.

Three things make this more than a leaderboard entry.

The winning residual objective splits by the same axis as the probe world. NPLM residual concatenation wins on fine-grained data (Cars 3/3, Flowers 2/3); plain softmax residual concatenation wins on texture and coarse data (DTD 3/3, Galaxy10 3/3). The estimator split from §4 reappears *inside* the residual construction, which we read as evidence that it is a property of what the objectives encode rather than of how they are deployed.

NPLM residual concatenations keep calibration while raising the probe. Cars/VISReg at probe 0.855 with per-event 0.263 and the best `eucl` and `mahaT` on Cars to date is **the first space in the campaign to hold both currencies at once**. Plain residual concatenations, by contrast, often buy their probe *with* calibration — Galaxy10/LeJEPA reaches probe 0.975 at per-event 0.001, which is a warning label, not a win.

The effect is essentially deterministic. Multi-seed checks give Cars/VISReg paired $\Delta = +0.148 \pm 0.004$ (3/3 seeds) and CIFAR-10 paired $\Delta = +0.0157 \pm 0.0004$ (5/5 seeds).

And the records survive replication. Every dataset record in this paper has now been re-run over three paired seeds, so we can cite them with uncertainties rather than as single draws:

cell	construction	record
Aircraft/VISReg	`supcon-ft`→res-nplm concat	0.866 ± 0.002
Galaxy10/LeJEPA	`supcon-ft`→res concat	0.971 ± 0.004
Flowers/DINO	**three-way** res concat (no discovery)	$\mathbf{0.910 \pm 0.012}$
DTD/VISReg	res concat + discovery	0.867 ± 0.004
Cars/VISReg	**width-matched** res concat	$\mathbf{0.884 \pm 0.003}$

The shift is milder than we expected and not uniformly downward: the archived Aircraft and DTD numbers were the *low* draws of their triples, and Galaxy10 moved by 0.004. Only Flowers falls materially (-0.019), and it is also the widest spread in the campaign — its pre-discovery concatenation ranges 0.787–0.885 across seeds, with the discovery step compensating for weak parents ($+0.106$ on the weakest seed against $+0.021$ on the strongest). That is worth noting as a mechanism in its own right: discovery's apparent benefit is partly variance reduction — and iterating the residual turns out to do the same job better, stabilising Flowers from $\mathrm{sd} = 0.040$ to 0.012 while setting a record (0.910 ± 0.012) that beats the discovery-assisted number *without any discovery step*. **No ordering flipped**, so every comparative claim in this paper stands; but one objective ranking elsewhere (plain-res vs res-nplm on CIFAR-100) did flip under averaging, which is why we report spreads throughout.

Stacking is purity-gated. Running discovery on top of the residual winners stacks only where the residual space already separates the novelty: Flowers $+0.021$ (new record 0.906) and DTD/VISReg $+0.015$ (new record 0.862, finally beating `simclr-ft`), flat or slightly negative elsewhere. On fine-grained data the discovery fine-tune erodes exactly the residual concatenation's probe directions.

7 Scale-free statistics see what distances miss

Every calibration statistic in §3 is a distance. Distances fail in a specific and predictable way: when novel classes sit *on* the data manifold, near some seen centroid, minimum-distance scores are blind by construction. This is the fine-grained regime — exactly the regime where pooling fails (§6).

We therefore add a *scale-free* statistic: the Levina–Bickel estimator of local intrinsic dimension,

$$\mathrm{LID}_k(\mathbf{x}) \;=\; -\left[\frac{1}{k-1}\sum_{j=1}^{k-1}\log\frac{r_j(\mathbf{x})}{r_k(\mathbf{x})}\right]^{-1}, \tag{4}$$

computed against seen-class training references ($k = 20$, max 4000 refs). It depends only on *ratios* of neighbor distances, so it is invariant to the scale deformations that break both fixed-σ kernels and distance thresholds. Higher LID indicates novelty: holdout LID exceeds seen LID in nearly every space we measured — novelty lives in locally higher-dimensional neighborhoods.

Result On Flowers-102, LID reaches AUC $= 0.951$–0.955 using *no novelty labels at all*, exceeding the best supervised probe on that dataset (0.906). This is the only unsupervised score to beat a trained probe anywhere in the campaign, and it is a new Flowers novelty record.

Controls We treat a result of this shape with suspicion and ran the corresponding controls. *Holdout rotation*: over 10 random 10-class holdouts on a frozen trunk, Flowers LID gives 0.879 ± 0.018 (min 0.844) versus a kNN-distance baseline at 0.692 ± 0.047 — classes 92–101 are not special. *Distance baselines*: LID beats kNN-distance by $+0.08$–0.12 and `eucl` by $+0.03$–0.06 everywhere on Flowers, with $\mathrm{Spearman}(\mathrm{LID}, k\mathrm{NN}\text{-}d) \;=\; 0.55$–$0.92$ — correlated, but carrying independent signal. *Robustness*: bootstrap 95% CIs ± 0.006; per-holdout-class AUC minimum 0.86–0.92 (no single-class driver); k-sensitivity plateaus at $k \in [20, 50]$ and degrades to ≈ 0.78 at $k = 5$ (the ratio estimator needs neighbors). A rank ensemble with `eucl` adds nothing — LID subsumes it.

And where it fails The same protocol on Cars gives 0.545 ± 0.040, and on VISReg fine-tuned Cars spaces the ordering flips: kNN-distance and `eucl` beat LID. Aircraft and Galaxy10 sit at 0.56–0.66. LID**'s advantage is regime-specific, and the regime is a dataset property, not a protocol artifact** — the identical rotation protocol produces 0.88 on Flowers and 0.55 on Cars. The reading: Flowers holdouts sit on the flower manifold with small distances and mediocre distance-based scores, but a distinct local dimensional signature; Cars holdouts do not have one.

The signature is class-level, not within-class We initially read "distinct local dimensional signature" as within-class structure, and predicted that LID computed on the *residual* half of a concatenation — where within-class variation lives by construction — would be stronger than on the parent half. It is not. On Flowers the residual half is uniformly *worse* (0.923 vs 0.951 on DINO, 0.825 vs 0.893 on LeJEPA, 0.733 vs 0.954 on VISReg), with the concatenation tracking the parent. The Flowers LID signal lives in the parent, class-structure half: **the novelty is a class-level dimensional anomaly, not a within-class one**. On Cars the prediction half-holds — residual-half LID beats parent on DINO (0.677 vs 0.629) and VISReg (0.752 vs 0.682, the best Cars LID on record) but not LeJEPA — and `eucl` at 0.747 still matches it, so there is no regime flip: Cars remains a distance-score dataset.

The mechanism, finally measured Decomposing LID by neighbourhood composition settles what "class-level" means, and it is neither of the two stories we had. Splitting each holdout point's 20 neighbours at the modal reference class, the *same-class* component carries essentially all of the discrimination on Flowers (0.94–0.95 against 0.95 for the full statistic) while the *mixed* component sits at chance (0.52–0.62) — the opposite of our prediction that between-class structure would carry the signal. A novel flower looks like a member of one particular seen class by distance, but its radial-ratio profile *within that class's own neighbourhood* is wrong: it sits off the class's local sheet. Between-class distances carry nothing. Sobering coda: the bare neighbourhood composition — simply the fraction of a point's neighbours that are not of the modal class — scores 0.83–0.94, nearly matching LID itself, so most of what our ratio estimator provides is available from a far simpler statistic. On Cars no component separates (0.54–0.73), consistent with everything else about that dataset.

Composition detects; radial ratios regime-select Two follow-ups sharpen what LID is contributing. First, a parameter-free alternative — the bare fraction of a point's neighbours not of the modal class — ties LID on the on-manifold cells (within 0.016–0.033 on Flowers, and beating it on CIFAR-10, 0.865 against 0.672), so on those cells the ratio estimator is not earning its complexity. But it does not replace LID, for a specific and instructive reason: on Aircraft composition *anti*-correlates with novelty (0.42–0.48), because novel variants sit in high-purity same-manufacturer neighbourhoods and counting therefore votes "member". LID holds above chance there, and a holdout-rotation control retains a real LID edge on Flowers (0.875 ± 0.029 vs 0.818 ± 0.038). Second, when the two are used as *regime* predictors rather than scores, the roles invert: composition carries no regime signal at all (base rate), while the within-modal-class radial-ratio dispersion — precisely the quantity LID inverts — identifies the regime in $15/17$ cells. The division of labour is clean: composition detects, radial ratios regime-select.

But the regime is not predictable out of sample That $15/17$ was in-sample: the threshold separating the two regimes was chosen on the same 17 cells it was then scored on, and a Spearman of 0.37 says the relation is threshold-like rather than monotone. We therefore ran the test that distinguishes a description from a predictor, with the order of operations enforced: the threshold was frozen on the original cells and written to disk, predictions on genuinely held-out cells were committed *before* any scoring, and only then was the outcome measured.

It fails. On three held-out Food-101 cells the frozen predictor scores $2/3$ against a pre-registered bar of 80% — and the more damning detail is that it returned the *same* answer for all three inputs, so it made no discriminating call at all. We had committed in advance to a stopping rule for this outcome, and we apply it: **the radial-ratio feature is a descriptive correlation, not a predictor, and we make no further attempts.** What the paper may claim is that the regime is label-free identifiable *in-sample*, and no more.

One mitigating fact, which we record without leaning on it: all three held-out gaps satisfy $|\Delta\mathrm{AUC}| \leq 0.03$, so Food-101 sits on the regime boundary where the sign is nearly arbitrary and where the choice of score barely matters anyway. A held-out cell with a large true gap would have been more diagnostic. But accepting that as an excuse after seeing the result is precisely what pre-registration exists to prevent, so the verdict stands.

We could not predict the regime in advance The obvious next step is to turn "which regime am I in" into a measurement, since the mechanism above — holdouts on- versus off-manifold — looks label-free. We tried three candidate diagnostics across 17 champion cells (pool distance-ratio, TwoNN intrinsic-dimension gap between tail and seen population, and an LID gap) and **none of them tracks the measured LID$-$`eucl` AUC gap**: Spearman $+0.07$, $+0.09$, -0.16, with a best sign-accuracy of $12/17$ against an $11/17$ base rate for always predicting LID. The LID-versus-distance regime rule therefore remains *empirical*: we can state which datasets fall on which side and give a mechanism, but we cannot currently tell a practitioner which side they are on without novelty labels. We report this because the negative result is more useful than the method we hoped to deliver.

That same sweep did strengthen LID's standing where it works. LID wins the gap in $12/17$ champion cells, and on the CIFAR-100 res-nplm concatenation the margin is decisive — LID 0.738 against `eucl` 0.356, a $+0.382$ gap that makes **LID the only usable novelty score in that space** — with galaxy10/LeJEPA at $+0.273$. Aircraft sits firmly on the other side (-0.11 to -0.20).

LID **as a pool scorer** Because LID is scale-free it should also fix the failure mode where a discovery fine-tune inflates the space and the distance-based pool threshold decays. It does: substituting LID for min-anchor-distance as the pool scorer holds round-2 purity at 0.60–0.68 where the distance

space (CIFAR-100, 100-D)	agree@1	agree@5	dendro purity	silhouette
`supcon` (parent)	**0.700**	**0.468**	0.591	0.197
`supcon_sigreg`	0.690	0.460	**0.603**	**0.199**
`supcon`→res-nplm (concat)	0.670	0.452	0.564	0.188
`supcon`→res (concat)	0.640	0.406	0.514	0.162
`simclr`	0.570	0.366	0.457	0.118
`nplm_bilinear`	0.530	0.334	0.417	0.059
`supcon`→res (residual half)	0.490	0.288	0.365	0.029
`lejepa`	0.450	0.342	0.427	0.046

Table 12: Semantic structure against the CIFAR-100 coarse-label partition (chance agree@1 $= 0.040$). Semantics track *supervision*, not probe quality and not calibration quality.

scorer collapses to 0.18–0.33, a $\sim 3\times$ improvement, and *strictly dominates* as a pooling mechanism. But the downstream battery does not move (probe deltas $+0.000$, -0.029, $+0.007$). Above the purity gate, purity is not the binding constraint — the fine-tune dynamics are. This is a clean example of why a battery, rather than a single number, is necessary: a mechanism can be strictly better and operationally irrelevant at the same time, and only a multi-currency report shows both facts.

8 Interpretability, measured

A discovery that cannot be explained is not yet a discovery. We treat semantic legibility as a measurable property and find it dissociates from the other currencies as sharply as they dissociate from each other.

The founding question Are otters near beavers? On a CIFAR-100 supervised parent space with `beaver` held out entirely — never labeled, never in the corpus — the nearest-centroid table reads `otter`→`seal`(0.02), `beaver`*(0.08), `bear`(0.11) and `beaver`* →`otter`(0.08), `shrew`(0.09), `seal`(0.10), `porcupine`(0.13). The never-labeled holdout lands mutually top-2 with otter purely from fine-tuning on the other 99 classes. The same holds on Galaxy10: the never-labeled `edge-on bulge` sits 0.06 from `edge-on no bulge`, its true sibling, with `cigar smooth` next — which is what an edge-on disk looks like. Open-world novelty settles where it semantically belongs, and this is the property that makes a discovered cluster explainable.

Semantics track supervision, not the probe The extreme case is Galaxy10 on DINO: `supcon-ft` recovers the four-group morphology partition *perfectly* (dendrogram purity 1.000, agree@1 0.9), while the label-free `nplm-bil-ft` — despite its calibration strengths — has essentially no semantic structure (purity 0.49, silhouette -0.08). The probe-versus-calibration dissociation is *also* a semantic-structure dissociation, and this is the strongest argument in the paper for reporting Currency IV separately: a practitioner selecting on calibration alone can land on a space in which no discovered cluster can be described.

Concatenation is semantically free Residual concatenations inherit the parent's semantics exactly (Galaxy10 res-nplm concat purity 1.000, equal to parent) while adding the calibration half. On CIFAR-100 the record residual concatenation *tightens* the semantic ring around the held-out beaver (0.06–0.10) at near-parent purity. Meanwhile the residual half *alone* scrambles semantics by design (beaver → mushroom/crab/lobster, purity 0.365) — semantics ride in the parent half, and the concatenation restores them while keeping the probe gain. This is a strong practical argument for concatenation over replacement.

What "fine-grained mistakes" actually are On Cars, the space organizes by body style as much as by make (make-agree@1 0.40 against chance 0.04), with errors like Audi convertible → BMW convertible. These are not failures; they are visually coherent groupings that a make-based partition scores as wrong. We note this because superclass agreement metrics can penalize exactly the structure a scientist would want.

metric	ideal	measures	fails when
r_{LL}	1	corr(proxy $-\frac{1}{2}d^2$, true $\log\mathcal{N}(z;\boldsymbol{\mu}_c,\Sigma_c)$)	shape is wrong
slope	1	OLS slope of true on proxy; empirically $\approx 1/\sigma^2$	width $\neq 1$
r_{LLR}	1	same, for pairwise differences (constants cancel)	test geometry is wrong
ECE	0	calibration of $\mathrm{softmax}_c(-\frac{1}{2}d^2)$ as a posterior	distance is not believable
SW	1	sliced-Wasserstein Gaussianity of labelled components	components non-Gaussian
RMS	1	per-dimension component width, i.e. the fitted σ	width $\neq 1$

Table 13: The interpretability panel. Reference density is a per-class Gaussian with shrunk full covariance. Report class separation $\mathrm{sep} = d_{\min}/\overline{\mathrm{RMS}}$ alongside: a space can be perfectly faithful and completely undiscriminating, so interpretability and discrimination are separate axes.

synthetic space	r_{LL}	slope	r_{LLR}	ECE	SW	RMS
ideal $\mathcal{N}(\boldsymbol{\mu}, I)$	0.988	**1.012**	0.988	**0.000**	**0.98**	**1.00**
width $2\times$	0.986	0.267	0.982	0.041	0.97	2.00
width $0.5\times$	0.989	4.282	0.990	0.000	1.02	0.50
anisotropic $4\times$	0.634	0.399	0.780	0.073	1.01	1.41
heavy tails t_3	0.951	1.146	0.899	0.008	13.71	0.98

Table 14: Panel validation on synthetic spaces with known departures. The ideal space returns the ideal values; **slope** recovers $1/\sigma^2$ exactly (0.267 at width $2\times$, 4.28 at $0.5\times$); anisotropy is the one departure that degrades r_{LL} rather than the scale; and heavy tails drive SW to 13.7 while leaving slope near 1 — confirming that shape and scale are independent failure modes, and that neither number substitutes for the other.

8.1 Is distance a log-likelihood? A six-number panel

The interpretability results above are qualitative: nearest-centroid tables a domain expert can read, and superclass agreement against a partition. They do not answer the question the program is actually built around, which is sharper and fully quantitative:

Is the distance from a point to a labelled anchor a Δ log-likelihood in a hypothesis test?

This is the ideal situation for a discovery instrument. If it holds, a distance is not a heuristic score but a statement about relative likelihood — readable by a domain expert, and simultaneously the correct quantity for a Neyman–Pearson test. It is also exactly what the SIGReg marginal is supposed to deliver, because if class c is $\mathcal{N}(\boldsymbol{\mu}_c, I)$ then $\log p(z \mid c) = -\frac{1}{2}\|z-\boldsymbol{\mu}_c\|^2 + \text{const}$ with the *same* constant for every class, so for two hypotheses

$$\Delta \log L \;=\; -\tfrac{1}{2}\Big(\|z-\boldsymbol{\mu}_c\|^2 - \|z-\boldsymbol{\mu}_{c'}\|^2\Big), \tag{5}$$

i.e. the squared-distance difference *is* twice the log-likelihood ratio, with no free scale. Departure from Eq. (5) is measurable, and we propose six numbers that measure it.

Two design points are worth drawing out. First, r_{LL} alone is *not* sufficient and would have been misleading on its own: a space at twice the correct width still scores $r_{\mathrm{LL}} = 0.986$ while its slope is 0.267, meaning a unit of $\frac{1}{2}d^2$ buys a quarter of a unit of log-likelihood. Correlation says the ordering is right; slope says whether the *units* are. A discovery threshold set on Eq. (5) depends on the latter. Second, SW and slope move independently — heavy-tailed components can have exactly the right width — so the Gaussianity of the labelled components and the calibration of their width are genuinely separate questions, and the panel reports both.

8.2 The panel's verdict: no space achieves unit width

We ran the panel across the campaign. One caveat first, because it bounds what can be claimed: the transfer cells have 10–40 samples per class in 100-D, so the per-class covariance is shrinkage-dominated (condition numbers 10^6–10^{13}) and their panel values are qualitative only. The CIFAR

dataset	space	$r_{\rm LLR}$	slope	ECE	RMS	sep
CIFAR-10	`supcon`	0.68	—	0.033	0.45	7.7
CIFAR-10	`res` (child)	0.81	4.3	—	**0.79**	0.85
CIFAR-10	`res-nplm` (child)	—	—	—	0.13	—
CIFAR-10	`res-cat`	**0.82**	—	**0.048**	—	5.5
CIFAR-10	`res-nplm-cat`	0.67	—	0.035	—	**10.6**
CIFAR-100	`supcon`	0.27	—	0.16	—	—
CIFAR-100	`res` (child)	0.64	—	0.51	—	—

Table 15: The panel on the quantitative (CIFAR) cells. Ideal: $r_{\rm LLR}$ and RMS $= 1$, ECE $= 0$, sep large.

cells, with $450+$ samples per class, are the quantitative ones. This is a limitation of the datasets, not of the panel, but it means the sharpest statements below are CIFAR statements.

The headline is a negative one about the program's own premise. **No space reaches** slope ≈ 1**, because no space achieves unit class-conditional width.** Fitted widths run 0.13–0.79, *all narrower* than the target, so raw squared distances overstate $\Delta \log L$ by a factor of 4 to 37. The SIGReg marginal pins the *aggregate* distribution to $\mathcal{N}(0, I)$ and the campaign has been reading that as though it pinned the class conditionals; it does not, and the gap is large. A practitioner converting a distance into a likelihood ratio via Eq. (5) would be overconfident by more than an order of magnitude. The per-class second-moment constraint that would fix this is exactly what the classwise marginal was supposed to supply — and §4.2 shows it never reaches its anchors either.

Two further readings. First, **the panel found the corner it was designed to find**: the plain-residual child on CIFAR-10 is the most faithful space in the campaign ($r_{\rm LLR} = 0.81$, RMS 0.79, the closest anything comes to unit width) and simultaneously the *least* discriminative (sep $= 0.85$, i.e. neighbouring classes sit less than one σ apart). Perfectly faithful and operationally useless. The concatenations are the deployable compromise — `res-cat` pairs $r_{\rm LLR} = 0.82$ with ECE 0.048 at sep $= 5.5$ — which is a quantitative version of the qualitative claim made throughout this paper that concatenation costs nothing.

Second, **interpretability degrades with class count**. CIFAR-100 tops out at $r_{\rm LLR} = 0.64$ and `supcon` there reaches only 0.27 with ECE 0.16: the distance-as-log-likelihood reading is weakest exactly where mahaT is weakest (§4). Interpretability and class count trade off, which is worth knowing before promising a domain expert that distances in a many-class embedding can be read as likelihood ratios.

Auditing our own headline Because the width claim is central and the transfer cells are shrinkage-dominated, we audited it. Subsampling the well-estimated CIFAR cells to 10–40 samples per class — the transfer regime — leaves RMS stable within ± 0.05 of its full-sample value on every space, while $r_{\rm LLR}$ degrades sharply (CIFAR-100 `supcon` $0.28 \to 0.04$). PCA projections to 16/32-D, where the covariance is estimable, barely move $r_{\rm LLR}$, so the low transfer values are not a dimension artifact; and reporting under isotropic rather than full-covariance references raises them to 0.73–0.91, bracketing how much is unestimability. The verdict is that **the width result stands on every cell**, being a first-moment quantity, while the correlation and ECE columns on transfer cells should be cited only as [full, isotropic] bounds. We have restricted our quantitative claims accordingly.

On our own prediction We predicted that SIGReg and NPLM arms would sit near unit width with slope near 1, and that SupCon-family spaces would show slope far from 1. The second half is right and the first is wrong in an informative way: *every* space is too narrow, including the ones whose objective explicitly targets unit variance. The dissociation of §4 is visible in the panel, but so is a failure mode common to all arms that none of our previous metrics exposed.

8.3 How spaces relate to each other

Cross-space measurements supply mechanisms for several claims above.

Supervision erases the pretraining base; self-supervision preserves it. The same supervised arm fine-tuned from two *different* bases reaches CKA ≈ 0.74 on average — as similar as two *different supervised losses* on the same base. The label signal pulls every pretraining toward one shared

identity / defect	repair, and did it work?	what it bought
The summed critic gradient is the calibration residual, so poorly calibrated runs should be identifiable	yes — the residual is exactly that quantity	nothing: it does not track probe quality; selecting on it *underperforms* random seeds
A bare distance critic cannot satisfy $\mathbb{E}_{\mathrm{ref}}[e^g] = 1$	yes — adding the bias term drives the residual from 1.77 to 0.03	nothing: probe falls, per-event falls $0.042 \to 0.008$
The pairwise critic and the event-level test estimate the same object, so one should initialize the other	n/a — the transfer is real but harmful	worse: power $0.800 \to 0.560$ at fixed budget
If the test statistic is what we care about, optimize it directly	yes — it optimizes	worse on every column *including* the statistic optimized
SIGReg should give unit class-conditional width	yes — width 0.37–0.65 $\to$ 0.80–1.04	nothing for calibration: slope stalls, r_{LLR} flat-to-down, ECE worsens

Table 16: Five independent identity repairs. Every one succeeded as a repair. None produced the benefit that motivated it.

supervised geometry, which is precisely why the regime rules of §4.2 are base-invariant. Label-free `simclr-ft` is far less base-invariant (CKA 0.46–0.62). Consistently, on DTD — where `simclr` wins the probe — `simclr-ft`'s *local* structure transfers across bases better than the supervised arms' (mutual-kNN 0.28 vs 0.16): its texture neighborhoods are the transferable part.

NPLM spaces are reorganizations, not new information. The NPLM arms are geometric outliers everywhere (CKA 0.2–0.36, mutual-kNN 0.03–0.06 against every other arm, LLE weight-transfer ratio > 1) — yet they are highly *decodable from* softmax spaces (asymmetric ridge R^2 up to 0.97 at CKA 0.36). Calibration is a re-encoding of information the supervised space already contains. This explains why concatenation works so well and why a single loss struggles: the two currencies are different *codes* for overlapping content.

Residual children are mostly linear re-maps — and drift is capacity, not help. Parent-vs-child ridge R^2 is 0.97–0.98 on coarse data (the child is nearly a linear re-map; calibration is re-weighting) and 0.91–0.95 with mutual-kNN ≈ 0.2 on fine-grained data (locally reorganized, 5–10% genuinely new variance). Within the transfer grid the concatenation gains track the unexplained fraction. But CIFAR refutes the naive version of this rule: there children drift *much* further (R^2 down to 0.32) while producing the campaign's *smallest* probe gains (+0.006–0.016 vs Cars' $+0.148$). Unexplained variance is necessary and not sufficient — what pays is whether the new variance encodes the holdout.

Loss families have intrinsic-dimension fingerprints. TwoNN intrinsic dimension orders the arms identically in every cell: NPLM supervised most compressed (ID $\approx$ 2–3), SIGReg/SSL arms mid (5–7), softmax and frozen highest (9–13). The Gaussian target and the density-ratio collapse are directly visible as dimension, which makes ID a cheap objective-provenance diagnostic.

9 Repairing an identity is not the same as improving a space

The result we least expected from this campaign is methodological, and it is the one we would most want carried into other work. Over its course we identified five places where the program failed to satisfy a theoretical identity it was built around. In each case we could name the defect precisely, we designed a targeted repair, and **the repair worked on its own terms and delivered nothing downstream**.

We want to state this carefully, because the obvious over-reading is wrong. The repairs are not *harmful* as a class: the width term is safe at its operating point and is in fact a useful regularizer on our record concatenation (probe $0.9551 \to 0.9593$, separation $5.88 \to 10.93$), and two of the five were merely neutral. Nor is the theory wrong — each identity holds exactly where it was derived. The claim is narrower and, we think, more useful: **in this program, satisfying a distributional**

identity was never the operative cause of a space being good. What did move the numbers was architectural (residual decomposition and concatenation), procedural (freezing the space during discovery, matching the pool statistic to the novelty geometry), and — in the one case we found by accident — a continuous hyperparameter nobody had swept (§4.1).

The practical counsel is to treat a satisfied identity as a *license* rather than a *mechanism*. Pinning the marginal is what makes the clustering correctly specified and the Neyman–Pearson statistic well posed (§5.2); that is a real and load-bearing role. It is not a reason to expect the resulting space to detect better, and on our evidence it does not. Work that motivates an objective by an identity should measure the downstream quantity separately, and should be prepared for the two to come apart.

10 A reporting protocol

We distill the campaign into a protocol we would want applied to any embedding space advertised for scientific discovery.

1. **Report all four currencies.** Probe-only or geometry-only conclusions are, in our experience, wrong roughly as often as they are right, because the currencies are anti-correlated by construction.
2. **State the number of seen classes, the embedding dimension, and the centroid-to-anchor distance.** These determine which corner of the design cube is even available (classwise marginals require $\dim \geq n_{\text{classes}}$).
3. **Resample the holdout draw, not just the seed.** Which classes are held out matters *more* than which seed is used: across random single-class draws on CIFAR-100 the probe spread is $\mathrm{sd} \approx 0.019$ against a seed spread of 0.001–0.010, and mahaT ranges 0.391–0.569. Cross-holdout generalization claims need draw-resampled intervals, and calibration figures should be quoted with draw ranges. (Reassuringly, the conventional "last N classes" holdout sits inside the random-draw distribution, so it is not a confound — only an under-reported source of variance.)
4. **Never claim a probe gap below 0.02 from one seed.** Single-seed spread is ≈ 0.017. Two rankings in this campaign flipped under averaging.
5. **Use annealed median-heuristic bandwidths** for any kernel two-sample statistic compared across spaces with different scales. A fixed bandwidth reports zero power on a space that is fine.
6. **Report pool purity next to any discovery gain.** A gain at purity < 0.15 is refinement, not discovery, and behaves differently (it depends on substrate alignment, not on the pool).
7. **Include at least one scale-free statistic, and do not assume it agrees with your dataset-level test.** Distance scores are structurally blind to on-manifold novelty; LID costs nothing and, in one regime, beat every supervised alternative we had. But the per-event and kernel-test regimes are close to complementary (§5.6): Flowers is the campaign's best LID dataset and among its worst SparKer datasets. Report both; neither substitutes for the other. We cannot currently tell you in advance which regime you are in (§7).
8. **Before fine-tuning as part of discovery, check whether you need to.** On fine-grained data the discovery gain lives in the anchors and the update is close to pure cost (§5.8); freezing the space makes the step strictly non-negative and keeps the pool threshold valid into round two.
9. **Check the class-conditional width before reading distances as likelihoods.** Every space we measured is too narrow — fitted σ of 0.13–0.79 against a unit target — so squared distances overstate $\Delta \log L$ by 4–$37\times$ (§8.2). An aggregate Gaussian marginal does not deliver unit class conditionals, and the difference is an order of magnitude in the likelihood ratio.
10. **Establish whether your novelty is on-manifold or outlying** (§5.10) before choosing a score, a pool, or whether to fine-tune at all. One diagnostic decides six downstream choices.

11. **Sweep your inherited constants before believing a ceiling.** Two limits we reported as structural were artifacts of a knob nobody varied (§4.1, §5.10), and the best CIFAR-100 space we have was found by a configuration slip. A limit established by sweeping one family of methods is a limit on that family. Record which constants were measured and which were inherited.
12. **Audit semantics explicitly.** Superclass agreement, dendrogram purity, and a nearest-centroid table a domain expert can read. Calibration does not imply legibility — we have a space with strong calibration and silhouette -0.08.

And a default recipe For a practitioner who wants one starting point: fine-tune end-to-end with `supcon` (adding a global SIGReg marginal if the data are fine-grained and low-shot), then train a residual head on $\mathbf{z} - \boldsymbol{\mu}_y$ with an NPLM objective on fine-grained data or a softmax objective on coarse/texture data, and evaluate the concatenation. Add a discovery pass only if measured pool purity exceeds ≈ 0.3. Score with the four-currency battery, including LID.

11 Limitations

Our evidence is entirely in the image domain; the regime rules are stated in terms of class counts and samples per class and we expect them to transfer, but we have not tested them on the scientific modalities (event-level collider data, spectra, time series) that motivate the work. The "discovery" setting is simulated by holding out known classes, which is a proxy: real novelty is not a withheld member of a curated taxonomy, and the Cars result — where novel classes are not geometrically outlying — suggests the proxy's difficulty varies more than the datasets' nominal difficulty does. Several conclusions rest on 3–5 seeds, which resolves 0.02-scale effects but not smaller ones. Two claims in the paper rest on parametrizations we did not fully test: the calibration identity does not hold for the bias-free distance critic our implementation uses (App. A), so the distance-NPLM arms' detection strength is evidence of good scaling rather than of achieved calibration, and adding the bias term is untested. And we cannot predict the LID-versus-distance regime without labels (§7) — three candidate diagnostics failed — so that rule is a description of our datasets, not a tool. Our discovery loop is one specific algorithm (quantile pool, BIC k-selection, anchor merge, prototype fine-tune) and the purity-gating conclusion may be specific to it. A protocol limit surfaced late: under alphabetically-ordered holdouts the number of scorable superclasses can fall to single digits, so some absolute superclass-agreement figures rest on few classes and should be read as relative comparisons rather than absolute rates. Finally, the interpretability metrics require a superclass partition, which is available in our datasets and frequently is not in the settings we care about; the nearest-centroid tables remain available but require expert reading rather than a number.

12 Conclusion

Embedding spaces built for scientific discovery need to be evaluated as instruments, not as classifiers. When we do so — across 79 controlled experiments, seven datasets, three pretraining bases and ten objectives — the picture is consistent. The part we would most want carried forward is the unification of §5: a pipeline that discovers new classes need not bolt a clustering heuristic and an anomaly test onto a representation trained for something else. Fix the marginal, and the density-ratio objective that shapes the space is the statistic that certifies a discovery, while the clustering that proposes candidates is correctly specified under the same density. When the pipeline then fails, it fails for a reason the diagnostics name in advance — rate on CIFAR-100, tail geometry on Cars, fine-tune dynamics on Aircraft — which we value more than any single AUC. Alongside that: linear readability and calibrated geometry are anti-correlated properties that no single loss in our design space achieves together; the repair is architectural (residual decomposition and concatenation) rather than a better loss; scale-free statistics reach novelty that distance-based scores cannot see, in a regime one can identify in advance; and semantic legibility is a third, separately measurable axis that tracks supervision alone. The distributional-prior program that LeJEPA and SIGReg opened up is, we think, the right foundation — specifying the marginal is what makes any of these quantities well posed.

What it needs alongside it is a metrology: a standard battery, honest uncertainties, and the discipline to report the currency a space *lacks*.

Reproducibility statement

All experiments are scripted (`experiments/NN_*.py`, each with a mandatory `--quick` smoke mode) over a shared library (`supersig/`: losses, training loops, metrics, discovery). Result arrays are archived as `logs/expNN/*.npz` with fixed key conventions, condensed into per-dataset master tables under `docs/`. Seeds are fixed and identical across paired cells; the probe uses seeds $1000{+}s$. Known reproduction jitter from cuDNN nondeterminism is $\approx \pm 0.003$ in AUC and is stated wherever single-seed records are cited.

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A Gradient descent under NPLM versus InfoNCE, and the log-likelihood readout

The dissociation of §4 is stated empirically in the main text. This appendix derives it. We compare the two estimators of the same population object — the pointwise mutual information — at the level of their *gradients*, and show that essentially every qualitative difference reported in the campaign (the vanishing per-event power of softmax spaces, the inertness of λ, the heavy-tailed seed variance of the bilinear NPLM arm, the base-robustness of the distance NPLM arm, and the intrinsic-dimension fingerprints) follows from a single structural fact: **InfoNCE is flat along the direction that carries the absolute normalization, and NPLM is strictly convex along it.**

We flag scope honestly. The derivations in §A.2–§A.4 are analytical and new to this paper; the links drawn to campaign measurements in §A.5 are *post-hoc consistency checks* against data already collected, not preregistered tests. We indicate in each case what a decisive experiment would be.

A.1 Two estimators, one target

Fix a batch $\{(x_i, x_i^+)\}_{i=1}^N$ and write $g_{ij} = g_\theta(x_i, x_j^+)$, with the diagonal as positive pairs and the off-diagonal as the shuffled product-of-marginals reference. The two objectives are

$$\widehat{\mathcal{L}}_{\mathrm{NCE}} = -\frac{1}{N}\sum_i \Big[g_{ii} - \log \sum_j e^{g_{ij}} \Big], \tag{6}$$

$$\widehat{\mathcal{L}}_{\mathrm{NPLM}} = \frac{1}{N(N-1)} \sum_{i \neq j} \big(e^{g_{ij}} - 1 \big) - \frac{1}{N} \sum_i g_{ii}. \tag{7}$$

They share the batch geometry exactly; only the functional differs. Both are consistent estimators of PMI — but of different *gauge classes* of it, as the gradients make immediate.

A.2 The row-sum identity: a flat direction versus a convex one

Differentiating (6) with respect to the critic entries gives

$$\frac{\partial \widehat{\mathcal{L}}_{\mathrm{NCE}}}{\partial g_{ij}} = \frac{1}{N}\big(\sigma_{ij} - \delta_{ij}\big), \qquad \sigma_{ij} = \frac{e^{g_{ij}}}{\sum_k e^{g_{ik}}}, \tag{8}$$

while (7) gives

$$\frac{\partial \widehat{\mathcal{L}}_{\mathrm{NPLM}}}{\partial g_{ij}} = \begin{cases} \dfrac{e^{g_{ij}}}{N(N-1)}, & i \neq j \ \ \text{(reference)}, \\ -\dfrac{1}{N}, & i = j \ \ \text{(positive)}. \end{cases} \tag{9}$$

Three differences are visible at once.

(i) InfoNCE gradients are row-stochastic; NPLM's are not From (8), $\sum_j \partial \widehat{\mathcal{L}}_{\text{NCE}}/\partial g_{ij} = 0$ for *every* anchor i, because $\sigma_{i\cdot}$ is a probability vector. The InfoNCE gradient is therefore purely *relative*: it can reallocate mass among the entries of a row but can never change a row's total. Formally, consider shifting the critic by an arbitrary per-anchor function, $g_{ij} \mapsto g_{ij} + c_i$. Then σ_{ij} is unchanged and

$$\widehat{\mathcal{L}}_{\text{NCE}}[g + c] = \widehat{\mathcal{L}}_{\text{NCE}}[g] \quad \text{exactly, for all } c. \tag{10}$$

The per-anchor shift is an exact *flat direction* of the loss — zero gradient, zero curvature. This is the entire content of "InfoNCE recovers PMI up to a per-row constant": the constant is not merely unidentified, it is *invisible to gradient descent*, so no amount of training, data, or capacity can recover it.

For NPLM the same shift gives, with $\bar{c}$ a uniform shift for clarity,

$$\widehat{\mathcal{L}}_{\text{NPLM}}[g + \bar{c}] = e^{\bar{c}}\, \widehat{\mathbb{E}}_{\text{ref}}\big[e^g\big] - 1 - \widehat{\mathbb{E}}_{\text{pos}}[g] - \bar{c}, \tag{11}$$

so that

$$\frac{d\widehat{\mathcal{L}}_{\text{NPLM}}}{d\bar{c}} = e^{\bar{c}}\, \widehat{\mathbb{E}}_{\text{ref}}\big[e^g\big] - 1, \qquad \frac{d^2\widehat{\mathcal{L}}_{\text{NPLM}}}{d\bar{c}^2} = e^{\bar{c}}\, \widehat{\mathbb{E}}_{\text{ref}}\big[e^g\big] \;>\; 0. \tag{12}$$

The direction InfoNCE cannot see is *strictly convex* for NPLM, with a unique minimum at $\bar{c}^\star = -\log \widehat{\mathbb{E}}_{\text{ref}}[e^g]$, i.e. exactly at the calibration condition $\widehat{\mathbb{E}}_{\text{ref}}[e^{g^\star}] = 1$. Equivalently: summing (9) over all pairs,

$$\sum_{i,j} \frac{\partial \widehat{\mathcal{L}}_{\text{NPLM}}}{\partial g_{ij}} = \widehat{\mathbb{E}}_{\text{ref}}\big[e^g\big] - 1 = \text{(calibration residual)}. \tag{13}$$

The total gradient of the NPLM loss *is* the miscalibration. Gradient descent on (7) performs calibration as an inseparable part of fitting, and (13) is computable at zero cost during training.

We flag one negative result about how far that observation goes. It is tempting to use the residual as a *label-free model-selection* criterion — prefer the seed whose terminal residual is closest to zero — and we proposed exactly that. It does not work: over five paired seeds the rank correlation between $|\text{residual}|$ and probe is $+0.30$ and -0.10 on the two bilinear arms, and selecting the best-of-five by residual *underperforms* the random-seed mean (0.839 against 0.891). Part of the reason is structural — for distance critics the residual cannot reach zero at all, see below — but even for bilinear critics, where it can, per-seed variation in calibration does not track per-seed variation in probe quality. The seed variance appears to live in the balance between the interaction and marginal terms rather than in calibration failure. Eq. (13) remains a valid *diagnostic* of whether a run calibrated; it is not a predictor of how good the resulting space is.

This is the precise sense in which per-event detection power is *structurally* unavailable to a softmax objective. A per-event threshold is a statement about the absolute value of the score; (8) says that value is a gauge choice on which the objective exerts no force. The observed 0.000 per-event power of every SupCon-family space in the campaign is not a tuning failure or a weak baseline — it is this identity.

(ii) Why max-subtraction destroys calibration The standard numerical guard for softmax losses, $g_{ij} \mapsto g_{ij} - \max_k g_{ik}$, is a per-anchor shift. By (i) it is exactly free for InfoNCE and exactly *fatal* for NPLM: it moves the minimizer from $g^\star$ to $g^\star + c$, the one quantity the loss was chosen to pin. The correct guard clamps the exponent, $e^{\min(g,\, g_{\max})}$, which by (9) sets the gradient of pathological reference pairs to a constant instead of shifting the optimum, leaving $g^\star$ untouched wherever $g < g_{\max}$. The campaign uses $g_{\max} = 30$ and verifies the resulting critic against analytic PMI at correlation > 0.9 with mean offset ≈ 0; max-subtraction drives the offset to hundreds.

(iii) Bounded versus exponential gradients From (8), $|\partial \widehat{\mathcal{L}}_{\text{NCE}}/\partial g_{ij}| \le 1/N$ *unconditionally*. InfoNCE cannot produce a large critic gradient no matter how badly scaled the embedding is. From (9), the NPLM reference gradient is $e^{g_{ij}}/N(N-1)$, which is unbounded above. This single asymmetry accounts for the optimization folklore around NPLM, and we can quantify it.

A.3 Gradient variance and the two critics

Suppose the critic values on reference pairs are approximately Gaussian, $g \sim \mathcal{N}(\mu, s^2)$. Then the relative variance of the reference gradient is

$$\frac{\operatorname{Var}\left[e^g\right]}{\left(\mathbb{E}[e^g]\right)^2} = e^{s^2} - 1, \tag{14}$$

which grows *exponentially* in the spread of the critic, while the corresponding quantity for (8) is $O(1)$ because $\sigma_{ij} \in [0, 1]$. A modest critic spread of $s = 3$ already gives a relative gradient standard deviation of ≈ 90. NPLM training is therefore not merely "sometimes unstable": its gradient noise is controlled by a quantity, s, that is itself a free consequence of the parametrization.

This makes the *critic* axis of the design cube, which in the main text is one knob among four, the decisive one for optimization. Compare the two critics used in the campaign:

$$\text{distance:} \quad g = -\tfrac{1}{2}\|z - z'\|^2/\tau \;\leq\; 0 \qquad \Longrightarrow \qquad e^g \leq 1 \quad \text{(bounded by construction)}, \tag{15}$$

$$\text{bilinear:} \quad g = \langle z, z' \rangle/\tau \;\in\; \mathbb{R} \qquad \Longrightarrow \qquad e^g \text{ unbounded.} \tag{16}$$

The distance critic makes the exponential gradient *automatically bounded* — the pathology of (14) cannot occur, and the clamp never activates in the normal range. The bilinear critic has no such protection: a single well-aligned reference pair can dominate the batch gradient.

The same bound makes a bare distance critic uncalibratable The boundedness is not free, and we initially missed what it costs. If $g \leq 0$ everywhere then $e^g \leq 1$, hence

$$\mathbb{E}_{\text{ref}}\left[e^g\right] \;\leq\; 1 \qquad \text{with equality only in the degenerate case } g \equiv 0, \tag{17}$$

so the calibration residual of Eq. (13) is *structurally negative* and the condition $\mathbb{E}_{\text{ref}}[e^{g^\star}] = 1$ is **unreachable**. A bare distance critic cannot calibrate, however well it is optimized. The resolution is the bias term of the full parametrization, $g = -\frac{1}{2}\|z - z'\|^2 + b(x) + b(x')$, which restores the positive range; the implementation used throughout this campaign omits it. We therefore qualify the claim made in §5.1: the $e^g - 1$ term pins the absolute normalization *for the unconstrained functional and for critic families that can represent positive values* — the bilinear critic, or distance-plus-bias — but not for the bias-free distance critic as implemented. The distance-NPLM arms' strong per-event and kernel-test power (§5.5) is therefore evidence that their geometry is well-scaled, not that they achieve the calibration identity.

We added the bias, and calibration turned out not to be the lever The obvious repair is to restore the positive range with the bias term, and we ran it: distance-plus-bias, with and without a running normalizer, against the bilinear-plus-clamp reference over five paired seeds. The repair *works* in its own terms — the calibration residual falls from $|\cdot| = 1.77$ for the clamped reference to 0.03, so these variants genuinely satisfy $\mathbb{E}_{\text{ref}}[e^g] \approx 1$ where the bare distance critic provably cannot — and it buys **nothing**. Probe drops ($0.891 \pm 0.032 \to 0.776$–$0.819$), no variant cuts seed variance by the targeted factor at equal mean, and per-event power *falls* from 0.042 to 0.006–0.008: the falsifier we flagged as the one to watch, variance disappearing together with the absolute scale.

The honest conclusion is uncomfortable for the narrative this appendix develops. Calibration in the residual sense is neither the variance problem (§A.5) nor a performance lever; and the best label-free NPLM arm remains the *miscalibrated* bilinear-plus-clamp configuration, whose apparent defect is evidently where its probe performance lives. We therefore retain the structural claim — a bare distance critic cannot satisfy the identity, and that is worth knowing — while withdrawing the implication that fixing it should improve anything. The identity licenses the estimand; it does not by itself produce a better space.

	InfoNCE / SimCLR	NPLM
positive-pair gradient	$\frac{1}{N}(\sigma_{ii}-1)$, saturating	$-\frac{1}{N}$, constant
reference-pair gradient	$\frac{1}{N}\sigma_{ij} \in [0, \frac{1}{N}]$	$e^{g_{ij}}/N(N-1)$, unbounded
row sum of gradients	0 (exactly)	calibration residual, Eq. (13)
per-anchor shift $g \to g+c$	flat direction	strictly convex, Eq. (12)
recovered object	$\mathrm{PMI} + c(x)$	PMI, absolute
repulsion range	long (rank/softmax)	short (Gaussian kernel)
relative gradient variance	$O(1)$	$e^{s^2}-1$
stabilizer	max-subtraction (free)	exponent clamp (max-sub. is fatal)

Table 17: Gradient properties of the two estimators of the same population target. Every entry follows from (8)–(9).

A.4 The force law on the embedding

Push the gradients through to the embedding itself. Under the distance critic (the form the critic collapses to once SIGReg is active), $\partial g_{ij}/\partial z_i = -(z_i - z_j^+)$, and

$$\nabla_{z_i}\widehat{\mathcal{L}}_{\mathrm{NPLM}} = \underbrace{\frac{1}{N}\big(z_i - z_i^+\big)}_{\text{spring to positive}} \; - \; \underbrace{\frac{1}{N(N-1)}\sum_{j\neq i} e^{-\frac{1}{2}\|z_i - z_j^+\|^2}\big(z_i - z_j^+\big)}_{\text{repulsion}} \,. \tag{18}$$

Two features of (18) are worth stating plainly.

The repulsion is the gradient of a kernel density estimate. The weight $e^{-\frac{1}{2}\|z_i - z_j\|^2}$ is an unnormalized isotropic Gaussian kernel, so the second term is precisely $\nabla_{z_i}\hat{q}(z_i)$ for $\hat{q}$ a Gaussian KDE of the reference population. NPLM gradient descent is therefore: *pull toward your positive with a linear spring, and descend the estimated reference density*. The unit-width Gaussian supplies an intrinsic **length scale**, and that length scale is what makes distances absolute. A point far from all references feels essentially *no* repulsion — the force vanishes as $e^{-d^2/2}$ — so the space does not expend capacity separating what is already separated.

InfoNCE has no length scale. The corresponding repulsion weight is σ_{ij}, which is normalized *within a row*: even if every negative is far away, the row still sums to one and the repulsion is redistributed rather than extinguished. Contrastive repulsion is thus long-range and rank-based, and its fixed point is uniformity on the sphere rather than any particular scale. This is the geometric statement behind the main text's observation that `simclr` attains probe 0.867 with `eucl` 0.236: the objective arranges *angles* correctly and never had an opinion about radii.

The positive term does not saturate. In (8) the positive gradient is $\frac{1}{N}(\sigma_{ii}-1)$, which $\to 0$ as the positive is separated: InfoNCE anneals itself. In (9) it is the constant $-\frac{1}{N}$. NPLM pulls positives together with undiminished force forever, and equilibrium is reached *only* when the exponential repulsion balances it — that is, at the calibration condition, not at a separation criterion.

A.5 What this predicts, and what the campaign measured

The derivation yields five predictions. All five have a counterpart in the campaign data. We stress again that these are consistency checks on existing measurements, not independent confirmations.

1. *Softmax spaces have exactly zero transferable per-event power, regardless of probe quality.* Predicted by the flat direction in (i). Measured: per-event 0.000 for every SupCon-family space on every dataset and every signal fraction, including spaces at probe 0.94.

2. λ *is inert early;* τ *is not.* At initialization with a bilinear critic on raw (unnormalized) embeddings, g has spread of order tens, so by (9) the interaction gradient is of order e^g — numerically $\sim 10^{10}$ against $\sim 10^{-2}$ for $\lambda \cdot \mathrm{SIGReg}$. The marginal term cannot compete until the exponent has contracted, which takes ≈ 2 epochs; λ then shapes only the endgame. By contrast τ rescales the exponent itself and hence enters the gradient multiplicatively at all times.

Measured: λ ablations are flat, $\tau = 1$ is a sharp optimum on CIFAR-100 with $\tau \in \{0.5, 2\}$ each costing 0.06–0.10 probe.

3. *The bilinear NPLM arm should be the high-variance one; the distance NPLM arm should be the stable one.* Predicted by §A.3: $e^g \leq 1$ automatically for the distance critic, unbounded for bilinear. Measured, and to our mind the appendix's most striking correspondence: `nplm_bilinear` on CIFAR-100 gives 0.855 ± 0.042 over paired seeds, a spread $10\times$ that of `supcon` (0.940 ± 0.004), with a heavy right tail that produced a single-seed 0.935 we initially mistook for parity; whereas `nplm_sup_dist` is *the most base-robust arm in the campaign*, probe 0.72–0.73 across all three pretraining trunks (spread 0.009, versus `supcon`'s 0.076). The unexplained seed variance flagged in §4 is, on this account, the exponential gradient variance of (14) — and the prediction is that it is a property of the *critic*, not of the estimator.

   We have since run the decisive test — estimator held at NPLM, critic swept against positives at matched τ, five paired seeds — and it confirms the attribution while correcting our proposed diagnostic. Probe standard deviation splits $3.1\times$ by critic (bilinear 0.055, distance 0.018) against only $1.6\times$ by positives, and `dist-sup` is the stable corner at 0.8366 ± 0.0125. But the quantity we suggested logging, the raw critic spread $s = \mathrm{sd}(g)$, *anti*-correlates with the probe spread (-0.80), while the spread of e^g correlates $+0.80$ and separates the arms by two orders of magnitude (0.1–0.3 for distance critics, 12–35 for bilinear). The reason is the one-sidedness noted above: Eq. (14) assumed g approximately Gaussian, and a distance critic is bounded above, so it can carry a large $\mathrm{sd}(g)$ over strongly negative values while e^g stays squashed near zero. **The theory holds in its exponential form only**; practitioners should monitor $\mathrm{sd}(e^g)$, not $\mathrm{sd}(g)$.

4. *NPLM spaces should be compressed relative to softmax spaces.* Predicted by the short-range force law: distant pairs exert no force, and the non-saturating positive pull is balanced only at the calibration scale. Measured: TwoNN intrinsic dimension orders the arms identically in every cell of the grid — supervised NPLM $\approx$ 2–3, SIGReg/SSL 5–7, softmax and frozen 9–13.

5. *NPLM should reorganize rather than add information.* The Bregman identity for (7) states that the constrained minimizer is the projection of PMI onto the critic family in the divergence generated by $\phi(t) = e^t$; a different projection of the same target is still a function of that target. Measured: NPLM spaces are geometric outliers against every other arm (CKA 0.2–0.36, mutual-kNN 0.03–0.06) yet are highly *decodable from* softmax spaces (ridge R^2 up to 0.97 at CKA 0.36).

A.6 Why the log-likelihood readout needs both terms

Finally, the gradient view sharpens why neither objective alone delivers an interpretable likelihood surface, which is the motivation for the hybrid used throughout the campaign.

NPLM fixes the interaction and its absolute scale, but the critic family $g = \langle z, z' \rangle + b(x) + b(x')$ retains a gauge,

$$z \mapsto Rz + a, \qquad b(x) \mapsto b(x) - \langle a, z \rangle - \tfrac{1}{2}\|a\|^2 + \beta, \tag{19}$$

under which g, and hence the loss and *every gradient*, is exactly invariant. So NPLM alone determines what the pair interaction is but not how it is split between the embedding and the bias — the map into "log-likelihood space" is undetermined by descent. SIGReg removes precisely this residual freedom: pinning $p(z) = \mathcal{N}(0, I)$ kills the shift a (leaving only an irrelevant rotation R) and identifies the bias as the marginal log-density, $b_\theta(x) \to -\frac{1}{2}\|z\|^2 + \text{const}$. Substituting back collapses the critic to the pure distance form and yields the three quadratic identities

$$\log p(x) = -\tfrac{1}{2}\|z\|^2 + c_0, \qquad \log \frac{p(x, x')}{p(x)p(x')} = \langle z, z' \rangle, \qquad \log p(x, x') = -\tfrac{1}{2}\|z - z'\|^2 + c_1. \tag{20}$$

The division of labour is exact and is a statement about descent directions: **NPLM supplies curvature along the normalization direction that InfoNCE leaves flat; SIGReg supplies curvature along the gauge directions that NPLM leaves flat.** Each term is the other's null space.

Two caveats bound the claim, and both are visible in the main text's results. The identities hold exactly only when PMI lies in the bilinear-plus-separable family; otherwise the Bregman projection returns the nearest such surface, and the residual is real — it is one reading of why concatenating a softmax half with an NPLM half outperforms either alone (§6), since the two halves are different projections of overlapping content. And the readout degrades under collapse: at small embedding dimension and small τ the supervised optimum is a simplex equiangular tight frame in which all class centres are equidistant and the likelihood information is destroyed. The campaign's realizability condition ($\dim \geq n_{\text{classes}}$ for classwise anchors, §4.2) is the practical form of that caveat, and the observed regression of the bilinear NPLM arms at 64-D on CIFAR-100 — losing *both* currencies — is the kind of failure it predicts.

B Discovery reach: the fraction at which SparKer power crosses 0.95

Power reported at a fixed signal fraction answers "how well does this space do on my grid"; a search wants the inverse quantity — *how small a signal can this space find?* We therefore invert every archived power curve into a single sensitivity number,

$$f_{95} \;=\; \min\{\, f \;:\; \mathrm{power}_{\mathrm{SparKer}}(f) \;\geq\; 0.95\,\}, \tag{21}$$

the injected fraction at which the test becomes reliable. Lower is better. We take the first upward crossing and interpolate linearly in $\log f$, power curves being sigmoidal in $\log f$.

The measurement is limited by the fraction grid, not by the spaces This is the first thing to report, and it governs how the table should be read. Power is estimated from 50 signal toys, so it is quantized in steps of 0.02 and carries a Clopper–Pearson uncertainty of roughly ± 0.03 near 0.95. More seriously, the transfer grid is $f \in \{0.003, 0.01, 0.02, 0.05, 0.1\}$, and essentially every crossing falls in the final interval $[0.05, 0.1]$, across which power typically rises from ~ 0.1 to ~ 1.0. Of 106 newly covered spaces, 46 cross 0.95 inside the grid at all, and **only four have a crossing that is not pinned to that top interval**. We mark the rest with $*$ and treat them as "somewhere in $(0.05, 0.1)$" rather than as measurements. The cleanly determined values are CIFAR-10 `supcon` ($f_{95} = 0.019$) and `res-nplm` concat (0.023), Galaxy10/VISReg `supcon-ft-resnplm-cat` (0.046), and Cars/VISReg `supcon-ft-resnplm` (0.049); CIFAR-10 is the only dataset whose grid $(\ldots, 0.02, 0.03, 0.1)$ brackets its own crossing, which is precisely why its numbers are the trustworthy ones.

What the table nonetheless shows Three statements survive the caveat. First, **Flowers never reaches** 0.95 **in any of 54 archived series on any base**, peaking at $0.70/0.94/0.42$ on DINO/VISReg/LeJEPA. Combined with Flowers' campaign-best LID score of 0.951, this is the sharpest form of the regime complementarity of §5.6: the dataset whose novelty is easiest to detect per event is the one whose novelty is hardest to detect in aggregate. Second, **every best-in-cell space is a residual or concat construction** except on CIFAR-10, extending §5.6 from power-at-a-fraction to reach. Third, reach tracks the base ordering already established: VISReg gives the two cleanest crossings, LeJEPA is worst in every dataset where it appears ($1/6$, $1/6$, $0/8$ spaces crossing).

The dense scan, and the resulting citable ordering We subsequently ran the dense scan — $f \in \{0.02, \ldots, 0.10\}$ on each cell's champion — and it converts half the champion table from bounds into measurements: $8/17$ cells are now cleanly bracketed, against $4/106$ on the coarse grid. The resulting sensitivity ordering is citable:

CIFAR-100 0.029 < CIFAR-10 0.038 < Cars/VISReg 0.063 < Galaxy10/VISReg 0.067 <
Galaxy10/LeJEPA 0.071
< Aircraft/VISReg 0.072 < DTD/VISReg 0.079 ≈ Cars/DINO 0.079

with Clopper–Pearson bands of ± 0.002–0.008. Our prediction that every starred cell would resolve within 0.02 holds only partially: seven cells genuinely require $f > 0.1$, so for them the bound *is* the reach and not a grid artifact. Flowers is confirmed > 0.1 on all three bases with notably shallow

dataset	base	spaces crossing	best f_{95}	best space
CIFAR-10	—	5/5	**0.019**	`supcon`
CIFAR-100	—	5/5	0.047^*	`res-nplm-cat`
Galaxy10	VISReg	6/6	**0.046**	`supcon-ft-resnplm-cat`
Galaxy10	LeJEPA	6/6	0.082^*	`ss-ft-res-cat`
Galaxy10	DINO	1/6	0.099^*	`supcon-ft-resnplm`
Cars	VISReg	4/6	**0.049**	`supcon-ft-resnplm`
Cars	DINO	5/6	0.084^*	`supcon-ft-resnplm`
Cars	LeJEPA	1/6	0.099^*	`supcon-ft-resnplm-cat`
DTD	VISReg	2/6	0.091^*	`supcon-ft-res-cat`
DTD	DINO	4/6	0.092^*	`supcon-ft-resnplm-cat`
DTD	LeJEPA	1/6	0.096^*	`ss-ft-res-cat`
Aircraft	VISReg	4/8	0.082^*	`supcon-ft-resnplm`
Aircraft	DINO	2/8	0.097^*	`supcon-ft-resnplm`
Aircraft	LeJEPA	0/8	> 0.1	—
Flowers	DINO	0/6	> 0.1	— (max power 0.70)
Flowers	VISReg	0/6	> 0.1	— (max power 0.94)
Flowers	LeJEPA	0/6	> 0.1	— (max power 0.42)

Table 18: SparKer discovery reach by cell, over the 106 spaces of §5.6. $*$ marks a crossing pinned to the top grid interval $[0.05, 0.1]$ and therefore poorly determined. Bold marks the four cleanly bracketed values in the whole set.

curves (0.14–0.76 at $f = 0.1$) — the strongest form of the score-versus-statistic split, since the same dataset supports a 0.95 unsupervised per-event score.

C Full metric tables

Tables 23–27 then give the same metrics for the *residual* constructions — parent, residual half and concatenation for each objective — which the body discusses but which were previously only summarized by their best cell.

Tables 19–22 give every Part-A metric for every arm and base on the four end-to-end transfer datasets, so that the selective comparisons in the body can be checked against the complete grid. Values are single-seed unless a spread is quoted; per §3, probe gaps below 0.02 should not be read as real.

arm	probe (pre)	probe (post)	acc	supAUC	`eucl`	mahaT	mahaPC	f_{95} (SpK)
dino								
`supcon`	0.733	0.767	0.462	0.974	0.603	0.564	0.668	0.097*
`ss`	0.719	0.747	0.330	0.966	0.551	0.547	0.599	0.094*
`simclr`	0.624	0.684	0.108	0.776	0.469	0.476	0.535	>0.1
`sigreg-ssl`	0.584	0.690	0.082	0.803	0.483	0.496	0.526	>0.1
`nplm-bil`	0.591	0.704	0.063	0.760	0.506	0.496	0.511	>0.1
`nplm-sup`	0.549	0.600	0.105	0.916	0.555	0.547	0.550	>0.1
lejepa								
`supcon`	0.699	0.726	0.498	0.987	0.627	0.572	0.678	0.098*
`ss`	0.709	0.699	0.299	0.966	0.574	0.581	0.589	>0.1
`simclr`	0.618	0.614	0.079	0.808	0.469	0.453	0.478	>0.1
`sigreg-ssl`	0.563	0.617	0.067	0.782	0.491	0.465	0.509	>0.1
`nplm-bil`	0.611	0.658	0.111	0.848	0.491	0.476	0.527	>0.1
`nplm-sup`	0.582	0.658	0.190	0.943	0.572	0.579	0.582	0.094*
visreg								
`supcon`	0.707	0.758	0.555	0.990	0.715	0.661	0.717	0.094*
`ss`	0.741	0.712	0.351	0.972	0.559	0.548	0.611	>0.1
`simclr`	0.610	0.657	0.074	0.751	0.472	0.462	0.490	>0.1
`sigreg-ssl`	0.574	0.634	0.064	0.760	0.493	0.463	0.484	>0.1
`nplm-bil`	0.576	0.624	0.081	0.826	0.537	0.490	0.523	>0.1
`nplm-sup`	0.579	0.549	0.186	0.950	0.578	0.578	0.584	0.088*

Table 19: Stanford Cars (196 classes, holdouts 186–195). All six end-to-end fine-tuning arms on all three pretraining bases. Probe is holdout-novelty AUC (3 seeds), pre and post discovery; `acc` is nearest-centroid top-1 over seen classes; supAUC the prototype-posterior macro OvR AUC; `eucl`/mahaT/mahaPC are novelty AUCs of min-anchor distance and tied / per-class Mahalanobis. f_{95} is the SparKer discovery reach of App. B (lower is better; $*$ = crossing lies in the top grid interval and is poorly determined; > 0.1 = never reaches power 0.95 within the measured grid).

arm	probe (pre)	probe (post)	acc	supAUC	`eucl`	mahaT	mahaPC	f_{95} (SpK)
dino								
`supcon`	0.708	0.814	0.944	1.000	0.911	0.873	0.833	>0.1
`ss`	0.795	0.788	0.940	1.000	0.933	0.883	0.822	>0.1
`simclr`	0.743	0.754	0.839	0.996	0.670	0.673	0.722	>0.1
`sigreg-ssl`	0.732	0.777	0.799	0.994	0.710	0.706	0.744	>0.1
`nplm-bil`	0.664	0.808	0.531	0.947	0.667	0.646	0.644	>0.1
`nplm-sup`	0.625	0.624	0.518	0.975	0.659	0.668	0.667	>0.1
lejepa								
`supcon`	0.678	0.752	0.934	0.997	0.897	0.873	0.797	>0.1
`ss`	0.792	0.613	0.879	0.998	0.883	0.899	0.821	>0.1
`simclr`	0.725	0.740	0.717	0.980	0.582	0.584	0.689	>0.1
`sigreg-ssl`	0.684	0.719	0.589	0.975	0.677	0.738	0.688	>0.1
`nplm-bil`	0.768	0.722	0.685	0.981	0.711	0.741	0.717	>0.1
`nplm-sup`	0.734	0.706	0.663	0.984	0.751	0.808	0.772	>0.1
visreg								
`supcon`	0.713	0.810	0.957	1.000	0.949	0.919	0.879	>0.1
`ss`	0.754	0.743	0.911	0.999	0.902	0.895	0.839	>0.1
`simclr`	0.746	0.728	0.689	0.983	0.569	0.554	0.688	>0.1
`sigreg-ssl`	0.738	0.734	0.614	0.976	0.666	0.709	0.695	>0.1
`nplm-bil`	0.737	0.737	0.471	0.945	0.646	0.672	0.609	>0.1
`nplm-sup`	0.631	0.576	0.621	0.982	0.723	0.778	0.784	>0.1

Table 20: Flowers-102 (102 classes, holdouts 92–101, $\sim$18 imgs/class). All six end-to-end fine-tuning arms on all three pretraining bases. Probe is holdout-novelty AUC (3 seeds), pre and post discovery; `acc` is nearest-centroid top-1 over seen classes; supAUC the prototype-posterior macro OvR AUC; `eucl`/mahaT/mahaPC are novelty AUCs of min-anchor distance and tied / per-class Mahalanobis. f_{95} is the SparKer discovery reach of App. B (lower is better; $*$ = crossing lies in the top grid interval and is poorly determined; > 0.1 = never reaches power 0.95 within the measured grid).

arm	probe (pre)	probe (post)	acc	supAUC	`eucl`	mahaT	mahaPC	f_{95} (SpK)
dino								
`supcon`	0.799	0.811	0.759	0.980	0.704	0.646	0.639	0.096*
`ss`	0.782	0.785	0.782	0.984	0.801	0.742	0.704	$>$0.1
`simclr`	0.808	0.801	0.653	0.957	0.469	0.506	0.716	0.099*
`sigreg-ssl`	0.803	0.802	0.634	0.967	0.543	0.577	0.744	$>$0.1
`nplm-bil`	0.681	0.785	0.366	0.849	0.465	0.479	0.506	$>$0.1
`nplm-sup`	0.586	0.781	0.432	0.933	0.590	0.615	0.636	$>$0.1
lejepa								
`supcon`	0.755	0.790	0.747	0.982	0.706	0.732	0.711	$>$0.1
`ss`	0.787	0.784	0.784	0.987	0.804	0.786	0.774	0.098*
`simclr`	0.817	0.821	0.646	0.968	0.516	0.580	0.785	$>$0.1
`sigreg-ssl`	0.809	0.826	0.598	0.963	0.607	0.714	0.784	0.097*
`nplm-bil`	0.746	0.745	0.505	0.934	0.595	0.637	0.705	$>$0.1
`nplm-sup`	0.581	0.765	0.649	0.961	0.714	0.754	0.760	$>$0.1
visreg								
`supcon`	0.750	0.791	0.774	0.984	0.780	0.765	0.755	0.098*
`ss`	0.784	0.785	0.799	0.984	0.834	0.807	0.791	$>$0.1
`simclr`	0.854	0.848	0.626	0.946	0.453	0.517	0.730	$>$0.1
`sigreg-ssl`	0.808	0.790	0.607	0.963	0.601	0.727	0.797	0.095*
`nplm-bil`	0.778	0.811	0.551	0.942	0.601	0.629	0.719	$>$0.1
`nplm-sup`	0.569	0.707	0.597	0.956	0.718	0.743	0.751	0.099*

Table 21: DTD (47 classes, holdouts 37–46). All six end-to-end fine-tuning arms on all three pretraining bases. Probe is holdout-novelty AUC (3 seeds), pre and post discovery; `acc` is nearest-centroid top-1 over seen classes; supAUC the prototype-posterior macro OvR AUC; `eucl`/mahaT/mahaPC are novelty AUCs of min-anchor distance and tied / per-class Mahalanobis. f_{95} is the SparKer discovery reach of App. B (lower is better; $*$ = crossing lies in the top grid interval and is poorly determined; > 0.1 = never reaches power 0.95 within the measured grid).

arm	probe (pre)	probe (post)	acc	supAUC	`eucl`	mahaT	mahaPC	f_{95} (SpK)
dino								
`supcon`	0.938	0.939	0.610	0.909	0.666	0.786	0.687	>0.1
`ss`	0.872	0.926	0.558	0.890	0.633	0.691	0.631	>0.1
`simclr`	0.868	0.942	0.421	0.830	0.595	0.729	0.602	>0.1
`sigreg-ssl`	0.914	0.931	0.408	0.825	0.514	0.764	0.694	0.092*
`nplm-bil`	0.839	0.928	0.319	0.705	0.664	0.712	0.657	>0.1
`nplm-sup`	0.878	0.912	0.414	0.837	0.779	0.824	0.782	0.071*
lejepa								
`supcon`	0.919	0.931	0.750	0.955	0.646	0.722	0.693	0.084*
`ss`	0.803	0.871	0.704	0.938	0.668	0.751	0.762	0.082*
`simclr`	0.902	0.946	0.500	0.864	0.626	0.685	0.654	0.096*
`sigreg-ssl`	0.927	0.940	0.426	0.834	0.301	0.774	0.708	0.095*
`nplm-bil`	0.892	0.905	0.505	0.863	0.431	0.620	0.663	0.095*
`nplm-sup`	0.899	0.897	0.640	0.900	0.787	0.860	0.774	0.095*
visreg								
`supcon`	0.939	0.944	0.764	0.945	0.720	0.854	0.804	0.047
`ss`	0.865	0.945	0.729	0.943	0.650	0.710	0.724	0.078*
`simclr`	0.933	0.922	0.526	0.867	0.554	0.731	0.674	0.095*
`sigreg-ssl`	0.931	0.940	0.405	0.824	0.274	0.678	0.603	0.092*
`nplm-bil`	0.802	0.904	0.332	0.758	0.588	0.684	0.704	>0.1
`nplm-sup`	0.883	0.933	0.581	0.875	0.770	0.834	0.759	0.088*

Table 22: Galaxy10 DECals (10 classes, holdout $\{9\}$). All six end-to-end fine-tuning arms on all three pretraining bases. Probe is holdout-novelty AUC (3 seeds), pre and post discovery; `acc` is nearest-centroid top-1 over seen classes; supAUC the prototype-posterior macro OvR AUC; `eucl`/mahaT/mahaPC are novelty AUCs of min-anchor distance and tied / per-class Mahalanobis. f_{95} is the SparKer discovery reach of App. B (lower is better; $*$ = crossing lies in the top grid interval and is poorly determined; > 0.1 = never reaches power 0.95 within the measured grid).

space	probe	acc	`eucl`	mahaT	per-evt	f_{95}
dino						
`supcon-ft` (parent)	0.761	0.457	0.698	0.659	0.177	>0.1
`supcon-ft`→`res` (residual)	0.733	0.216	0.540	0.582	0.105	>0.1
`supcon-ft`→`res` (concat)	0.781	0.427	0.653	0.642	0.204	0.099*
`ss-ft` (parent)	0.802	0.418	0.500	0.604	0.024	>0.1
`ss-ft`→`res` (residual)	0.747	0.108	0.390	0.495	0.021	>0.1
`ss-ft`→`res` (concat)	0.816	0.348	0.431	0.591	0.024	>0.1
`supcon-ft`→`res-nplm` (residual)	0.811	0.392	0.728	0.673	0.267	0.097*
`supcon-ft`→`res-nplm` (concat)	0.816	0.455	0.714	0.667	0.225	>0.1
lejepa						
`supcon-ft` (parent)	0.798	0.561	0.718	0.723	0.255	>0.1
`supcon-ft`→`res` (residual)	0.719	0.250	0.398	0.504	0.018	>0.1
`supcon-ft`→`res` (concat)	0.814	0.328	0.419	0.690	0.027	>0.1
`ss-ft` (parent)	0.802	0.463	0.512	0.619	0.045	>0.1
`ss-ft`→`res` (residual)	0.724	0.158	0.409	0.471	0.012	>0.1
`ss-ft`→`res` (concat)	0.761	0.420	0.451	0.563	0.033	>0.1
`supcon-ft`→`res-nplm` (residual)	0.746	0.200	0.573	0.610	0.036	>0.1
`supcon-ft`→`res-nplm` (concat)	0.848	0.551	0.711	0.711	0.198	>0.1
visreg						
`supcon-ft` (parent)	0.809	0.596	0.731	0.720	0.249	0.096*
`supcon-ft`→`res` (residual)	0.792	0.314	0.419	0.504	0.009	>0.1
`supcon-ft`→`res` (concat)	0.839	0.544	0.601	0.659	0.141	0.096*
`ss-ft` (parent)	0.828	0.529	0.561	0.615	0.048	>0.1
`ss-ft`→`res` (residual)	0.750	0.131	0.398	0.475	0.021	>0.1
`ss-ft`→`res` (concat)	0.814	0.474	0.458	0.592	0.033	>0.1
`supcon-ft`→`res-nplm` (residual)	0.858	0.575	0.812	0.755	0.411	0.082*
`supcon-ft`→`res-nplm` (concat)	0.863	0.605	0.792	0.765	0.363	0.092*

Table 23: FGVC-Aircraft — residual constructions. Every parent, residual half and concatenation from the residual fine-tuning grid, on all three bases. The residual half alone is the entry to watch: it trades probe and nearest-centroid accuracy for calibration, and on the fine-grained cells it carries per-event power at or above its own parent (§5.6). f_{95} as in App. B.

space	probe	acc	`eucl`	mahaT	per-evt	f_{95}
dino						
`supcon-ft` (parent)	0.733	0.462	0.603	0.564	0.136	–
`supcon-ft`→`res` (residual)	0.767	0.329	0.551	0.512	0.143	0.090*
`supcon-ft`→`res` (concat)	0.819	0.466	0.603	0.538	0.143	0.095*
`ss-ft` (parent)	0.719	0.330	0.551	0.547	0.103	–
`ss-ft`→`res` (residual)	0.605	0.065	0.462	0.483	0.042	>0.1
`ss-ft`→`res` (concat)	0.749	0.269	0.506	0.538	0.049	0.099*
`supcon-ft`→`res-nplm` (residual)	0.787	0.423	0.547	0.538	0.143	0.084*
`supcon-ft`→`res-nplm` (concat)	0.821	0.486	0.603	0.548	0.148	0.094*
lejepa						
`supcon-ft` (parent)	0.699	0.498	0.627	0.572	0.124	–
`supcon-ft`→`res` (residual)	0.587	0.111	0.471	0.471	0.035	>0.1
`supcon-ft`→`res` (concat)	0.726	0.237	0.500	0.553	0.035	>0.1
`ss-ft` (parent)	0.709	0.299	0.574	0.581	0.056	–
`ss-ft`→`res` (residual)	0.627	0.090	0.449	0.440	0.031	>0.1
`ss-ft`→`res` (concat)	0.677	0.243	0.510	0.525	0.031	>0.1
`supcon-ft`→`res-nplm` (residual)	0.629	0.133	0.469	0.445	0.014	>0.1
`supcon-ft`→`res-nplm` (concat)	0.779	0.491	0.608	0.514	0.082	0.099*
visreg						
`supcon-ft` (parent)	0.707	0.555	0.715	0.661	0.223	–
`supcon-ft`→`res` (residual)	0.728	0.371	0.569	0.523	0.073	0.093*
`supcon-ft`→`res` (concat)	0.801	0.547	0.705	0.613	0.228	0.090*
`ss-ft` (parent)	0.741	0.351	0.559	0.548	0.056	–
`ss-ft`→`res` (residual)	0.587	0.081	0.437	0.448	0.026	>0.1
`ss-ft`→`res` (concat)	0.709	0.303	0.479	0.492	0.042	>0.1
`supcon-ft`→`res-nplm` (residual)	0.836	0.609	0.728	0.646	0.256	0.049
`supcon-ft`→`res-nplm` (concat)	0.855	0.608	0.747	0.678	0.263	0.089*

Table 24: Stanford Cars — residual constructions. Every parent, residual half and concatenation from the residual fine-tuning grid, on all three bases. The residual half alone is the entry to watch: it trades probe and nearest-centroid accuracy for calibration, and on the fine-grained cells it carries per-event power at or above its own parent (§5.6). f_{95} as in App. B.

space	probe	acc	`eucl`	mahaT	per-evt	f_{95}
dino						
`supcon-ft` (parent)	0.708	0.944	0.911	0.873	0.369	–
`supcon-ft`→`res` (residual)	0.808	0.696	0.633	0.568	0.089	>0.1
`supcon-ft`→`res` (concat)	0.876	0.912	0.882	0.834	0.278	>0.1
`ss-ft` (parent)	0.795	0.940	0.933	0.883	0.524	–
`ss-ft`→`res` (residual)	0.721	0.726	0.506	0.493	0.032	>0.1
`ss-ft`→`res` (concat)	0.823	0.913	0.874	0.829	0.239	>0.1
`supcon-ft`→`res-nplm` (residual)	0.839	0.896	0.828	0.784	0.229	>0.1
`supcon-ft`→`res-nplm` (concat)	0.885	0.936	0.897	0.843	0.334	>0.1
lejepa						
`supcon-ft` (parent)	0.678	0.934	0.897	0.873	0.334	–
`supcon-ft`→`res` (residual)	0.812	0.853	0.861	0.907	0.273	>0.1
`supcon-ft`→`res` (concat)	0.810	0.856	0.866	0.912	0.293	>0.1
`ss-ft` (parent)	0.792	0.879	0.883	0.899	0.386	–
`ss-ft`→`res` (residual)	0.772	0.746	0.760	0.790	0.150	>0.1
`ss-ft`→`res` (concat)	0.813	0.835	0.844	0.886	0.258	>0.1
`supcon-ft`→`res-nplm` (residual)	0.637	0.350	0.586	0.627	0.047	>0.1
`supcon-ft`→`res-nplm` (concat)	0.712	0.890	0.828	0.864	0.182	>0.1
visreg						
`supcon-ft` (parent)	0.713	0.957	0.949	0.919	0.675	–
`supcon-ft`→`res` (residual)	0.794	0.878	0.877	0.898	0.352	>0.1
`supcon-ft`→`res` (concat)	0.813	0.904	0.905	0.928	0.438	>0.1
`ss-ft` (parent)	0.754	0.911	0.902	0.895	0.472	–
`ss-ft`→`res` (residual)	0.688	0.735	0.755	0.780	0.162	>0.1
`ss-ft`→`res` (concat)	0.739	0.867	0.870	0.886	0.329	>0.1
`supcon-ft`→`res-nplm` (residual)	0.646	0.669	0.479	0.404	0.032	>0.1
`supcon-ft`→`res-nplm` (concat)	0.857	0.952	0.921	0.870	0.455	>0.1

Table 25: Flowers-102 — residual constructions. Every parent, residual half and concatenation from the residual fine-tuning grid, on all three bases. The residual half alone is the entry to watch: it trades probe and nearest-centroid accuracy for calibration, and on the fine-grained cells it carries per-event power at or above its own parent (§5.6). f_{95} as in App. B.

space	probe	acc	`eucl`	mahaT	per-evt	f_{95}
dino						
`supcon-ft` (parent)	0.799	0.759	0.704	0.646	0.040	–
`supcon-ft`→`res` (residual)	0.817	0.670	0.560	0.505	0.022	0.096*
`supcon-ft`→`res` (concat)	0.845	0.761	0.683	0.604	0.045	0.094*
`ss-ft` (parent)	0.782	0.782	0.801	0.742	0.207	–
`ss-ft`→`res` (residual)	0.781	0.621	0.503	0.498	0.033	>0.1
`ss-ft`→`res` (concat)	0.828	0.774	0.728	0.693	0.125	>0.1
`supcon-ft`→`res-nplm` (residual)	0.779	0.742	0.746	0.669	0.113	0.093*
`supcon-ft`→`res-nplm` (concat)	0.832	0.764	0.733	0.648	0.077	0.092*
lejepa						
`supcon-ft` (parent)	0.755	0.747	0.706	0.732	0.113	–
`supcon-ft`→`res` (residual)	0.789	0.697	0.658	0.711	0.050	>0.1
`supcon-ft`→`res` (concat)	0.826	0.714	0.676	0.766	0.050	>0.1
`ss-ft` (parent)	0.787	0.784	0.804	0.786	0.242	–
`ss-ft`→`res` (residual)	0.815	0.693	0.648	0.716	0.080	>0.1
`ss-ft`→`res` (concat)	0.831	0.759	0.763	0.791	0.158	0.096*
`supcon-ft`→`res-nplm` (residual)	0.732	0.547	0.532	0.536	0.060	>0.1
`supcon-ft`→`res-nplm` (concat)	0.799	0.743	0.661	0.698	0.110	>0.1
visreg						
`supcon-ft` (parent)	0.750	0.774	0.780	0.765	0.163	–
`supcon-ft`→`res` (residual)	0.802	0.688	0.528	0.587	0.007	>0.1
`supcon-ft`→`res` (concat)	0.847	0.772	0.716	0.758	0.077	0.091*
`ss-ft` (parent)	0.784	0.799	0.834	0.807	0.285	–
`ss-ft`→`res` (residual)	0.784	0.670	0.564	0.629	0.022	>0.1
`ss-ft`→`res` (concat)	0.840	0.786	0.770	0.792	0.188	>0.1
`supcon-ft`→`res-nplm` (residual)	0.778	0.763	0.718	0.685	0.098	>0.1
`supcon-ft`→`res-nplm` (concat)	0.818	0.780	0.777	0.750	0.128	0.098*

Table 26: DTD — residual constructions. Every parent, residual half and concatenation from the residual fine-tuning grid, on all three bases. The residual half alone is the entry to watch: it trades probe and nearest-centroid accuracy for calibration, and on the fine-grained cells it carries per-event power at or above its own parent (§5.6). f_{95} as in App. B.

space	probe	acc	`eucl`	mahaT	per-evt	f_{95}
dino						
`supcon-ft` (parent)	0.938	0.610	0.666	0.786	0.021	–
`supcon-ft`→`res` (residual)	0.952	0.536	0.465	0.673	0.018	>0.1
`supcon-ft`→`res` (concat)	0.955	0.589	0.559	0.762	0.021	>0.1
`ss-ft` (parent)	0.872	0.558	0.633	0.691	0.058	–
`ss-ft`→`res` (residual)	0.923	0.489	0.458	0.662	0.015	>0.1
`ss-ft`→`res` (concat)	0.930	0.529	0.525	0.694	0.041	>0.1
`supcon-ft`→`res-nplm` (residual)	0.925	0.585	0.775	0.807	0.281	0.099*
`supcon-ft`→`res-nplm` (concat)	0.947	0.611	0.702	0.802	0.091	>0.1
lejepa						
`supcon-ft` (parent)	0.919	0.750	0.646	0.722	0.007	–
`supcon-ft`→`res` (residual)	0.960	0.625	0.286	0.499	0.001	0.095*
`supcon-ft`→`res` (concat)	0.975	0.650	0.288	0.625	0.001	0.093*
`ss-ft` (parent)	0.803	0.704	0.668	0.751	0.054	–
`ss-ft`→`res` (residual)	0.854	0.599	0.352	0.598	0.008	0.096*
`ss-ft`→`res` (concat)	0.879	0.699	0.517	0.735	0.028	0.082*
`supcon-ft`→`res-nplm` (residual)	0.916	0.651	0.386	0.546	0.009	0.092*
`supcon-ft`→`res-nplm` (concat)	0.941	0.746	0.506	0.673	0.008	0.085*
visreg						
`supcon-ft` (parent)	0.939	0.764	0.720	0.854	0.311	–
`supcon-ft`→`res` (residual)	0.964	0.714	0.618	0.835	0.243	0.088*
`supcon-ft`→`res` (concat)	0.964	0.760	0.676	0.868	0.295	0.078*
`ss-ft` (parent)	0.865	0.729	0.650	0.710	0.069	–
`ss-ft`→`res` (residual)	0.916	0.635	0.311	0.586	0.015	0.095*
`ss-ft`→`res` (concat)	0.920	0.727	0.482	0.707	0.047	0.088*
`supcon-ft`→`res-nplm` (residual)	0.936	0.765	0.744	0.780	0.333	0.056*
`supcon-ft`→`res-nplm` (concat)	0.946	0.769	0.744	0.839	0.353	0.046

Table 27: Galaxy10 DECals — residual constructions. Every parent, residual half and concatenation from the residual fine-tuning grid, on all three bases. The residual half alone is the entry to watch: it trades probe and nearest-centroid accuracy for calibration, and on the fine-grained cells it carries per-event power at or above its own parent (§5.6). f_{95} as in App. B.